# CHAPTER

# 4

# Group Reporting III

## Accounting for Business Combinations and Non-controlling Interests under IFRS 3 in Post-acquisition Periods



## LEARNING OBJECTIVES

***After reading this chapter, you should be able to:***

**LO1** Understand the rationale for elimination of investment;

**LO2** Understand the concept of non-controlling interests;

**LO3** Appreciate the alternative measurement basis for non-controlling interests under IFRS 3;

**LO4** Know how to prepare consolidation journal entries relating to goodwill, depreciation, and amortization of differences between book values and fair values of identifiable assets, contingent liabilities of acquired subsidiaries, and non-controlling interests;

**LO5** Know how to prepare consolidation journal entries to allocate current and past income to non-controlling interests; and

**LO6** Understand the components of non-controlling interests and know how to analytically determine their balances.

# INTRODUCTION

In Chapter 3, we saw how the acquisition method requires the acquirer to recognize and measure the identifiable net assets of the acquiree at fair value, and also recognize the residual that is called goodwill. Although there are different forms of business combinations, they share the same features — there is an acquirer who gains control of one or more businesses. In substance, the acquisition of a subsidiary is effectively the acquisition of its net assets. Chapter 3 highlights the recognition and measurement principles that govern all types of business combinations. The acquirer should recognize and measure identifiable net assets at fair value. What is the impact of the fair value measurement in subsequent periods? This chapter focuses on the subsequent effects when the identifiable net assets are sold, consumed, extinguished, or amortized. Since the acquirer recognizes and measures the identifiable net assets at fair value, the sale, consumption, use, or settlement of the assets and liabilities of the acquiree should also be recorded at acquisition-date fair value. The acquirer also has to test for impairment of goodwill. In this chapter, we go deeper into the type of business combinations that result in the acquisition of subsidiaries, and show how consolidation journal entries are passed to record the subsequent effects of the acquisition.

The chapter explains accounting for non-controlling interests, and shows how the balance of non-controlling interests can be analyzed with respect to three components. The chapter illustrates consolidation journal entries to recognize non-controlling interests' share of equity.

The chapter also illustrates the accounting for business combinations in multiple periods. In consolidation, certain past consolidation adjusting entries have to be re-enacted to reflect adjustments that were passed in previous periods. This chapter explains the re-enactment process.

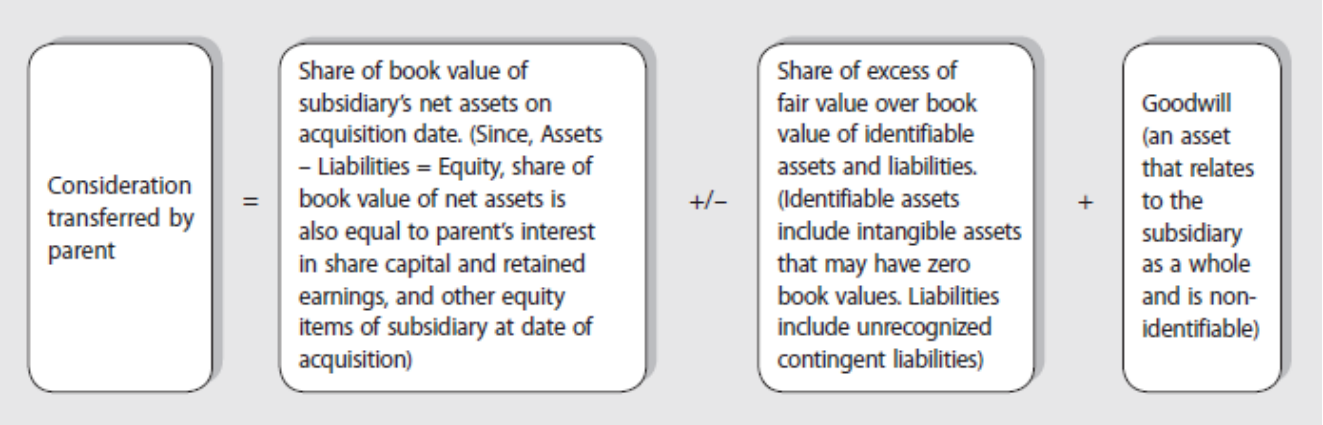
# ELIMINATION OF INVESTMENT IN A SUBSIDIARY

The first intragroup transaction that has to be eliminated in the consolidation worksheet is the investment by the parent in the subsidiary. Without the elimination, a double-counting of assets arises — the investment asset in the parent's statement of financial position is duplicated by the individual assets and the liabilities of the subsidiary acquired. By virtue of its power and ability to affect the variable returns of a subsidiary, the parent has control over the assets and liabilities of its subsidiary, and should, therefore, recognize the assets and liabilities of the subsidiary on the combined statement of financial position.

Hence, the consolidation process entails *asset substitution* whereby the investment in a subsidiary in the parent's separate financial statements is eliminated and substituted with the identifiable assets and liabilities of the subsidiary resulting in a residual called goodwill.

Consider what the parent is buying through its investment in a subsidiary. Essentially, the investment is a payment for three elements as shown in Figure 4.1.

**FIGURE 4.1 What a parent is paying for through its investment in a subsidiary**




In the group or consolidated financial statements, the investment account must be zero.

In other words, all items that have been recognized in the investment account must be "reversed" during the consolidation process and substituted with the three elements in Figure 4.1.

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## Why Eliminate the Investment Account?

An *asset substitution process* takes place whereby the investment account is replaced by the identifiable assets and liabilities of the subsidiary through the combination of the statement of financial position of the subsidiary with that of the parent. Hence, elimination of the investment account is necessary to avoid recognizing an asset in two forms.

On the equity side, an entity is not able to invest in itself; neither can it issue shares to itself. Share capital issued by a subsidiary has to be eliminated on consolidation. Further, any pre-acquisition retained earnings or pre-acquisition reserves of the subsidiary should not be included in consolidated equity. Pre-acquisition retained earnings arose prior to the acquisition of control by the parent, making it inappropriate to consolidate these earnings with the retained earnings of the parent. Hence, the investment in a subsidiary by a parent must be eliminated against the share capital, retained earnings, and other equity items of the subsidiary at the date of acquisition.

After eliminating the investment in a subsidiary against the share capital, retained earnings, and other equity items of the subsidiary that exist at the date of acquisition, a differential remains. As shown in Figure 4.1, the differential comprises two components, which are goodwill and the excess or deficit of fair value over book value of identifiable net assets of the acquired subsidiary.

## Re-enactment of the Elimination of Investment Entry

We have explained the basis for eliminating the investment in subsidiary account. In a subsequent year, the elimination of the investment has to be repeated because the parent's legal entity financial statements would include the investment in subsidiary balance. The elimination of investment entry has to be re-enacted for as long as the investment exists. The elimination process takes place in consolidation, where the building blocks are the legal entity financial statements. Consolidation journal entries are, therefore, important as they adjust the combined balances of the legal financial statements of parent and subsidiaries to arrive at proper consolidated balances. The consolidation worksheets, as introduced in Chapter 3, present the combination of the parent's and subsidiary's legal entity financial statements, and show the consolidation adjustments to arrive at the consolidated financial statements. We will see in this chapter that certain past consolidation adjustments have to be re-enacted in the current year to reflect the consolidated balances as at the beginning of the year.[1]

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The mechanics of consolidation requires the following three steps:

1. Showing the balances in the legal entity financial statements of the parent and the subsidiary in the consolidation worksheet;

2. Posting the consolidation journal entries in the worksheet; and
3. Summing up (1) and (2) to arrive at the consolidated balances.

Illustration 4.1 shows how the investment in a subsidiary of a parent is eliminated against the share capital and pre-acquisition retained earnings of the subsidiary. The pre-acquisition retained earnings are the retained earnings as at acquisition date. These retained earnings should not be included in the parent's consolidated financial statements as they arose before acquisition date and do not reflect the cumulative earnings of the new economic entity. The same principle applies to other components of equity such as revaluation reserves, foreign currency translation reserves, and other comprehensive income components. Under the acquisition method, the parent has to recognize the previously unrecognized intangible asset of the subsidiary in the consolidated financial statements. Deferred tax effects of the unrecognized intangible asset has to be also recognized. The residual is goodwill. In preparing the consolidated financial statements, the parent has to eliminate the investment at each reporting date. For as long as the investment remains in the books of the parent, the investment has to be eliminated on consolidation. In this example, the parent acquired a 100% interest. Later on in the chapter, we go deeper into the accounting for non-controlling interests where a parent's controlling interests are less than 100%.

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### ILLUSTRATION 4.1 Elimination of investment in subsidiary

Parent Co bought a 100% interest in Subsidiary Co for $200,000. At the date of acquisition, Subsidiary Co had the following equity balance:

| | |
|---|---|
| Share capital | $50,000 |
| Retained earnings | 30,000 |
| *Equity* | 80,000 |

At acquisition date, Subsidiary Co had an unrecognized intangible asset that had a fair value of $50,000. The tax rate at acquisition date was 20%.

In the consolidation worksheet below, we show the legal entity financial statements of Parent and Subsidiary. We then show the consolidation adjustments and the consolidated statement of financial position. You can see from the worksheets how the Investment in Subsidiary balance is eliminated against the share capital and retained earnings of Subsidiary Co. The investment does not offset the equity of the subsidiary perfectly and gives rise to a differential. The differential comprising of the intangible asset, deferred tax liability, and goodwill are recognized in the consolidated financial statements.

| | Parent | Subsidiary | Consolidation Adjustments Dr | Consolidation Adjustments Cr | Consolidated Statement of Financial Position |
|---|---|---|---|---|---|
| *Assets* | | | | | |
| Investment in Subsidiary | $200,000 | | | $200,000 | $ 0 |
| Goodwill (Note 2) | | | $ 80,000 | | 80,000 |
| Other net assets (Note 1) | 300,000 | $80,000 | 50,000 | 10,000 | 420,000 |
| | $500,000 | $80,000 | $130,000 | $210,000 | $500,000 |
| *Equity* | | | | | |
| Share capital | $100,000 | $50,000 | $ 50,000 | | $100,000 |
| Retained earnings | 400,000 | 30,000 | 30,000 | | 400,000 |
| | $500,000 | $80,000 | $ 80,000 | $ 0 | $500,000 |
| | | | $210,000 | $210,000 | |
| Note 1: | | | | | |
| Increase in other net assets due to recognition of intangible asset | | | | | $ 50,000 |
| Decrease in other net assets due to recognition of deferred tax liability | | | | | (10,000) |
| Net increase in other net assets | | | | | $ 40,000 |
| Note 2: Goodwill is the excess of the investment over the fair value of identifiable net assets: | | | | | |
| Investment in Subsidiary | | | $200,000 | | |
| Book value of equity or net assets | | | (80,000) | | |
| Fair value of intangible asset | | $50,000 | | | |
| Book value of intangible asset | | 0 | | | |
| Excess of fair value over book value | | $50,000 | | | |
| Deferred tax effects | | (10,000) | | | |
| | | | (40,000) | | |
| Goodwill | | | $ 80,000 | | |

The consolidation journal entry (CJE) that is passed is as follows:

*CJE1: Elimination of investment in subsidiary*

| | | |
|---|---|---|
| Dr Share capital | 50,000 | |
| Dr Retained earnings | 30,000 | |
| Dr Goodwill | 80,000 | |
| Dr Intangible asset | 50,000 | |
| Cr Investment in Subsidiary | | 200,000 |
| Cr Deferred tax liability | | 10,000 |
| | 210,000 | 210,000 |

CJE1 has to be re-enacted at each reporting date for as long as Parent has control over Subsidiary. Without CJE1, the investment is un-eliminated and the goodwill is unrecognized. We need to remember that the building blocks of the consolidation worksheet are the legal entity financial statements of the parent and its

subsidiaries. There are no carry-forward balances in consolidation, and each consolidation process is a fresh-start approach requiring the parent to re-enact certain consolidation adjustments.

# EFFECTS OF AMORTIZATION, DEPRECIATION, AND DISPOSAL OF UNDERVALUED OR OVERVALUED ASSETS AND LIABILITIES SUBSEQUENT TO ACQUISITION

In Chapter 3, we saw how business combinations can take different forms. An acquirer can acquire directly the business and recognize the identifiable net assets and goodwill on to its own legal entity financial statements. In this situation, the accounting is more straightforward. The acquirer will account for the acquired net assets in the same manner as assets purchased from third parties. If the acquired net assets included intangible assets, these would be recognized at fair value to reflect the exchange transaction. Subsequently, the acquirer would record the amortization on the basis of the initial fair value. In the direct acquisition of the net assets of a business, there is no need for the acquirer to prepare consolidation worksheets to show the effects of the acquisition. The legal entity financial statements of the acquirer would include the assets and liabilities of the enlarged entity. However, when the acquirer gains control of another entity, the acquirer has to consolidate the other entity to reflect the effects as if the acquirer had acquired the other entity's net assets. The acquirer would have to pass consolidation journal entries to recognize the acquisition-date fair values. Subsequently, the acquirer would have to pass consolidation journal entries to recognize the amortization of the acquisition-date fair values. The accounting is a little more complex than the situation when the acquirer acquires the net assets directly. Nonetheless, the effects in both types of business combinations must be the same.

From the discussion above, we note that IFRS 3 requires the following to be recognized:

1. Fair value of identifiable net assets of the acquiree as at acquisition date;
2. Intangible assets and contingent liabilities that meet the recognition criteria in IFRS 3, and that are not normally recognized in the separate financial statements of the acquiree;
3. Deferred tax liability or deferred tax asset arising from the above two items; and
4. Goodwill as a residual.

Since these adjustments are passed through the consolidated financial statements each year, the extinguishment of the fair value changes must also be reflected in the consolidated financial statements. From the group's perspective, subsequent extinguishment of assets and liabilities of the subsidiary must be determined based on the fair values recognized as at the date of acquisition. *Under the acquisition method, the acquirer is deemed to have acquired the identifiable net assets, both recognized and unrecognized, and the unidentifiable goodwill asset at fair values. The fair value of the assets of the acquiree at date of acquisition becomes the new "cost" to the group.* Subsequent amortization, depreciation, and cost of sales of the acquired assets are determined based on the fair value as at date of acquisition. The "set-up" entry (the elimination of the consideration transferred and the recognition of fair value adjustments) and the "amortization" entry (when the identifiable assets or liabilities are extinguished, disposed of, depreciated, amortized, or written off for impairment losses) must be repeated each year until the date of disposal of the investment in subsidiary or the date when control is lost. The process by which the fair value differences are extinguished is referred to as the "amortization" of the excess of the fair value over the book value of identifiable assets and liabilities.

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New internally-generated goodwill or subsequent appreciation in fair values are not recognized subsequent to acquisition date. The acquisition method recognizes fair value only at the critical event — the acquisition date. Subsequent to this date, the accounting for assets and liabilities is governed by the respective accounting standards. For example, research and development expenditures incurred after acquisition date will be subject to the recognition criteria in IAS 38 *Intangible Assets*. Research expenditures are expensed off and development expenditures are capitalized only if certain conditions are met.

Recall that the building blocks for consolidation are the separate financial statements of the parent and the subsidiary. Since the assets and liabilities on the subsidiary's statement of financial position are carried at book values, the "set-up" entry is necessary to adjust the book values to fair values. When an identifiable asset is disposed of or depreciated, a consolidation entry is required to adjust to acquisition-date fair value (FV), the cost of sales, or depreciation expense that is recorded at book value (BV) in the separate financial statements. Book value refers to the carrying amounts in the legal entity financial statements. Hence, in a period when the identifiable asset is sold or depreciated,

$$\text{BV of expense in separate financial statements} + \text{(FV – BV) adjustment to expense} = \text{FV of expense in consolidated financial statements}$$

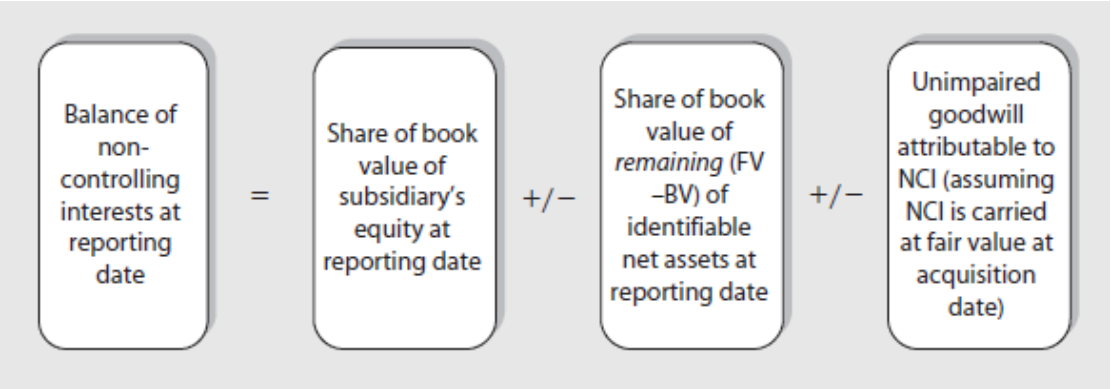
## Accounting for Non-controlling Interests under IFRS 3

The concept of non-controlling interests has been introduced in Chapter 3. Non-controlling interests (previously referred to as "minority interests" in the 2004 version of IFRS 3) arise only in the consolidated financial statements where one or more subsidiaries are not wholly owned by the parent. IFRS 10 Appendix A defines non-controlling interests as follows:

> Non-controlling interest is the equity in a subsidiary not attributable, directly or indirectly, to a parent.

In essence, non-controlling interests have a share in any change in equity or net assets of the acquired subsidiary. We can analyze the balance in non-controlling interests to the three components that we saw in Figure 4.1. While Figure 4.1 reflects the parent's share of equity in a subsidiary at acquisition date with respect to the three components, Figure 4.2 shows that a similar relationship applies to non-controlling interests' share in equity of a subsidiary at any reporting date. The balance of non-controlling interests changes from period to period; hence, we can analyze the balance to the three components at any time. Figure 4.2 shows the three components that are included in the balance of non-controlling interests at reporting date.

**FIGURE 4.2 Analysis of non-controlling interests**



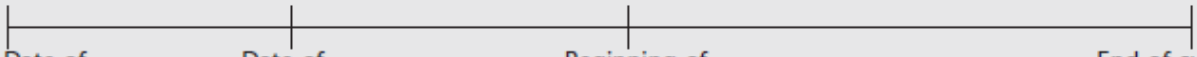

As we saw in Chapter 3, non-controlling interests can be measured in one of two ways. If non-controlling interests are measured at fair value at acquisition date, the third component will feature in the analysis. However, if non-controlling interests are measured as a proportion of identifiable net assets (excluding goodwill), the analysis will comprise only the first two components.

The analysis of non-controlling interests permits us to check our processes and gives us a quick way to assess the balance of non-controlling interests. However, another way of arriving at the non-controlling interests is to build up the balance chronologically through the consolidation process.

Figure 4.3 shows the reconstruction of non-controlling interests over time, beginning from acquisition date to the end of the current period.

**FIGURE 4.3 Reconstructing non-controlling interests on the statement of financial position**

Consider the following diagrammatic representation of the timeline that shows the components in the balance of non-controlling interests (NCI).



| Date of incorporation | Date of acquisition | Beginning of current year | End of current year |
|---|---|---|---|
| | NCI have a share of<br>1. Share capital<br>2. Retained earnings<br>3. Other equity<br>4. Fair value differentials<br>5. Goodwill | NCI have a share of<br>1. Change in share capital<br>2. Change in retained earnings<br>3. Change in other equity<br>4. Past amortization of fair value differential<br>5. Past impairment of goodwill | NCI have a share of<br>1. Profit after tax<br>2. Current amortization of fair value differential<br>3. Current impairment of goodwill<br>4. Dividends as a repayment of profits<br>5. Change in other equity |

At acquisition date, non-controlling interests would have been attributed the following three components if the fair value alternative is applied:

1. Book value or carrying amount of equity or net assets (that is, the amounts recognized in the subsidiary's legal entity financial statements);
2. Differences between the fair value and the book value or carrying amount of identifiable net assets (that is, the fair value differentials arising as at acquisition date); and
3. Goodwill (the unrecognized unidentifiable asset of the subsidiary).

You should take note that the first component attributed to non-controlling interests is the balance of equity (that is, share capital, retained earnings, other comprehensive income, and other equity components). These equity components would have arisen over the period from incorporation date to acquisition date. Hence, NCI as a collective body would have had an interest in the acquiree even before the acquisition date.



The timeline shows the continuous build-up of the balance of non-controlling interests. Unlike other accounts in the financial statements, the non-controlling interests do not exist in the legal books of the subsidiary. Non-controlling interests are equity interests in the economic entity and they feature in the consolidated financial statements only.

As no consolidation "ledgers" are kept, there are no "running balances" of non-controlling interests. The consolidation journal entries relating to non-controlling interests must be re-enacted each year to arrive at the current balance of non-controlling interests.

At each reporting date, the group will "re-create" the non-controlling interests account in the consolidated financial statements by recognizing the sequential build-up:

1. As of acquisition date;

2. From acquisition date to the beginning of the current period; and
3. During the current period.

Since the consolidation journal entries are repeated each year, we describe the process as a "re-enactment" of the attribution of equity to non-controlling interests.

Unlike a parent, which is entitled to consolidate retained earnings of a subsidiary from the date of acquisition, non-controlling interests as a collective body are entitled to their share of retained earnings of the subsidiary from incorporation. *There is no distinction between pre-acquisition and post-acquisition retained earnings for non-controlling interests*. The same applies to other comprehensive income; non-controlling interests collectively have a share of accumulated other comprehensive income arising from incorporation date to the current date. Hence, non-controlling interests have a share in the equity in a subsidiary at any given time when ownership in the subsidiary is held by parties other than the parent.

Because non-controlling interests have a share of the residual interest in the net assets of a subsidiary, they are normally a credit balance. Recall the accounting equation under the entity theory:

$$\text{Total equity (parent's and non-controlling interests')} = \text{Assets} - \text{Liabilities}$$

The above equation shows that equity on the consolidated statement of financial statement must include both the interests of equity owners of the parent and non-controlling interests of partially owned subsidiaries. IFRS 10 makes it clear that non-controlling interests are equity items and must be shown separately from the equity of the owners of the parent. On the other side of equation, total assets less total liabilities must be reported. Even if a subsidiary is only partly owned by the parent, the assets and liabilities of the subsidiary must be reported in full.

### Non-controlling Interests' Goodwill

In Chapter 2, we explain how the parent and entity theories determine the presentation and measurement of non-controlling interests in the consolidated financial statements. In Chapter 3, we also considered the alternative bases permitted for the measurement of non-controlling interests at acquisition date. IFRS 3 paragraph 19 allows non-controlling interests in the acquiree at acquisition date to be measured at either:

1. Fair value; or
2. As a proportion of the recognized amounts of the acquiree's identifiable net assets.

When the acquiree has both present ownership interests (that is, issued shares) and potential ordinary shares (for example, convertible bonds or options), the non-controlling interests' share of the potential ordinary shares must be measured at fair value. The option to measure non-controlling interests at fair value or a proportion of the recognized amounts of identifiable net assets of the acquiree applies only to the shares actually issued.



When non-controlling interests are measured at fair value, goodwill attributable to non-controlling interests will be recognized in the consolidated financial statements. Under the fair value basis, non-controlling interests comprise the following three components:

1. Share of book value of identifiable net assets of the subsidiary;
2. Share of (fair value – book value) of identifiable net assets of the subsidiary at acquisition date; and
3. Goodwill attributable to non-controlling interests at acquisition date.

Under the fair value basis, non-controlling interests are determined with reference to either active market prices of equity shares of the subsidiary at acquisition date or other valuation techniques (IFRS 3 App B:B44). Fair value per share of non-controlling interests may differ from fair value per share of the acquirer because a control premium may be paid by the acquirer (IFRS 3 App B:B45). For example, in a takeover bid, an acquirer may pay a 20% premium

over market price to gain control of the acquiree. The premium is included in goodwill paid for by the acquirer but is not proportionally attributable to non-controlling interests.

Under the second option where non-controlling interests are measured as a proportion of the acquiree's identifiable net assets, non-controlling interests comprise the following two components:

1. Share of book value of identifiable net assets of the subsidiary; and
2. Share of (fair value — book value) of identifiable net assets of the subsidiary at acquisition date.

Non-controlling interests' goodwill is not recognized under the second option under IFRS 3. The second option retains the treatment of non-controlling interests in IFRS 3 (2004).

The fair value basis permitted for the measurement of non-controlling interests at acquisition date brings revised IFRS 3 closer to the *entity theory*. The entity theory upholds consistency in accounting measurements of the assets and liabilities of the combined entity, regardless of who the stakeholders are. The entity theory does not support a fragmented valuation model as embodied in the parent theory, which focuses on information primarily for the shareholders of the parent company. The same accounting policy is applied to determining non-controlling interests' share of net assets as well as that of the group's. Hence, goodwill is recognized in full for the entity and fair value of identifiable assets and liabilities are valued in full to reflect both parent's and non-controlling interests' share. Although the acquisition of the subsidiary is entered into by the parent and not the non-controlling interests, IFRS 3 requires the fair value of the identifiable net assets of the subsidiary at date of acquisition to be recognized in full, as if the non-controlling interests were also a party to the acquisition.

However, the second alternative permitted by IFRS 3 with respect to recognizing non-controlling interests as a proportion of fair value of identifiable net assets and not full fair value diverges from the entity theory. In Chapter 3, we note that the Statement of Financial Accounting Standard (SFAS) 141R *Business Combinations* issued by the Financial Accounting Standards Board (FASB) in the US has chosen to adopt solely the fair value basis in the measurement of non-controlling interests. In permitting a choice, IFRS 3 is still some way from ensuring its objectives of comparability among IFRS-compliant companies and convergence with US GAAP.

Although the entity theory is not without its critics, it provides a conceptual basis to recognize, measure, and present non-controlling interests in a consistent manner. Unless otherwise indicated, the illustrations and end-of-chapter questions apply the fair value basis to accounting for non-controlling interests in line with the entity theory.



## Other Applications of the Entity Theory to Non-controlling Interests

The entity theory views all shareholders of the combined entity as equity holders. Under IFRS 10, non-controlling interests are to be presented as a component of equity in the consolidated financial statements. IFRS 10 paragraphs 22 and B94 (Appendix B) state:

> A parent shall present non-controlling interests in the consolidated statement of financial position within equity separately from the equity of the owners of the parent.
>
> An entity shall attribute the profit or loss and each component of other comprehensive income to the owners of the parent and to the non-controlling interests. The entity shall also attribute total comprehensive income to the owners of the parent and to the non-controlling interests even if this results in the non-controlling interests having a deficit balance.

Under the entity theory, the depreciation and amortization of fair value differentials that affect the parent also apply to the non-controlling interests. The chart in Figure 4.3 shows the build-up of non-controlling interests over time, beginning from acquisition date to the end of the current period.

## Recognizing Non-controlling Interests at Fair Value on Acquisition Date

IFRS 3 requires the recognition of non-controlling interests at either fair value or a proportion of identifiable net assets as at the date of acquisition. Hence, a consolidation entry is re-enacted each year to credit non-controlling

interests' at acquisition-date amount. In Illustration 4.1, we saw how the investment in subsidiary account is eliminated against share capital and pre-acquisition retained earnings of the subsidiary giving rise to goodwill and fair value differentials. If the parent owns less than 100% of the subsidiary, non-controlling interests have to be recognized. The recognition of non-controlling interests at the date of acquisition is normally integrated with the elimination of investment entry. Under the fair value basis, the entry is as follows:

Dr Share capital of subsidiary
Dr Retained earnings at date of acquisition
Dr Other equity of subsidiary at date of acquisition
Dr Goodwill (acquirer's and non-controlling interests' share)
Dr Identifiable net assets (excess of fair values over book values)
Dr/Cr Deferred tax asset/liability on fair value adjustments
  Cr Identifiable net assets (excess of book values over fair values)
  Cr Investment in subsidiary
  Cr Non-controlling interests (Fair value of non-controlling interests)

Under the alternative basis that recognizes non-controlling interests as a proportion of identifiable net assets, goodwill, and non-controlling interests recognized are generally smaller amounts than as recognized under the fair value basis:

Dr Share capital of subsidiary
Dr Retained earnings at date of acquisition
Dr Other equity of subsidiary at date of acquisition
Dr Goodwill (acquirer's goodwill only)
Dr Identifiable net assets (excess of fair values over book values)
Dr/Cr Deferred tax asset/liability on fair value adjustments
  Cr Identifiable net assets (excess of book values over fair values)
  Cr Investment in subsidiary
  Cr Non-controlling interests (NCI % × Fair value of identifiable net assets)



In Illustration 4.2, we show how the goodwill for non-controlling interests is calculated, if the fair value basis is applied.

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## ILLUSTRATION 4.2 Determining non-controlling interests' goodwill

The fair value of non-controlling interests that owned 10% of Subsidiary A as at 31 December 20x1 (acquisition date) was $25,000. The financial statements of Subsidiary A at acquisition date are shown below. On that date, Subsidiary A had unrecognized intangible assets that had a fair value of $40,000. Deferred tax liabilities should be recognized on the intangible assets at the tax rate of 20%. Determine non-controlling interests' goodwill as at acquisition date.

Subsidiary A
Statement of Financial Position
As at 31 December 20x1 (acquisition date)

| | | | |
|---|---|---|---|
| Net assets | $160,000 | | |
| Equity: | | | |
| Share capital | $140,000 | | |
| Retained earnings | 20,000 | | |
| | $160,000 | | |
| Fair value of non-controlling interests | | | $25,000 |
| Fair value of identifiable net assets: | | | |
| Book value of equity | | $160,000 | |
| Fair value of intangible assets | | 40,000 | |
| Deferred tax on intangible assets | | (8,000) | |
| | | $192,000 | |
| Non-controlling interests' share of fair value of identifiable net assets | | | 19,200 |
| Non-controlling interests' goodwill | | | $ 5,800 |

As with the nature of goodwill, non-controlling interests' goodwill should not include identifiable elements. Non-controlling interests' goodwill is the excess of fair value of non-controlling interests over non-controlling interests' share of fair value of identifiable net assets. The following equation shows the calculation of non-controlling interests' goodwill at acquisition date:

$$\text{Non-controlling interests' goodwill} = \text{Fair value of non-controlling interests} - (\text{Percentage of non-controlling interests} \times \text{Fair value of identifiable net assets})$$

Under the alternative basis, where non-controlling interests are measured as a proportion of identifiable net assets, non-controlling interests' goodwill is zero. The amount to be recognized as non-controlling interests is $19,200 that comprises non-controlling interests' share of identifiable net assets only.



## Allocation of the Changes in Equity from the Date of Acquisition to the Beginning of the Current Period

Non-controlling interests have a share of retained earnings and other equity over time. As stated earlier, there is no distinction between pre-acquisition or post-acquisition profits for non-controlling interests. The elimination of investment entry allocates pre-acquisition retained earnings, to non-controlling interests. Another entry, as shown below, is required to allocate the change in retained earnings from the date of acquisition to the beginning of the current period to non-controlling interests. After allocating non-controlling interests' share of both pre- and post-acquisition retained earnings to non-controlling interests, only the parent's share of the post-acquisition retained earnings of a partially-owned subsidiary remains. Group retained earnings comprises parent's retained earnings and post-acquisition retained earnings of subsidiaries attributable to parent's shareholders.

Dr Retained earnings (NCI % × Change in retained earnings from date of acquisition to beginning of current period)

Cr Non-controlling interests

Why is it necessary to debit retained earnings? Since the consolidation worksheet includes the full retained earnings of the subsidiary, this entry is necessary to transfer that portion of the subsidiary's retained earnings that belongs to non-controlling interests to non-controlling interests' equity in the statement of financial position.

### Allocation of Current Profit after Tax to Non-controlling Interests

Non-controlling interests are entitled to a share of **current profit after tax** of a subsidiary. The attribution or allocation entry is as follows:

Dr Income to non-controlling interests (income statement)

Cr Non-controlling interests (statement of financial position)

The same attribution principle applies to other comprehensive income. Non-controlling interests are attributed their share of other comprehensive income arising during a period. Other comprehensive income in IAS 1 *Presentation of Financial Statements* refers to revaluation surplus or deficit on property, plant, and equipment and intangible assets, re-measurement gains or losses on defined benefit plans, foreign currency translation reserves on net investments, fair value gains or losses on fair value through other comprehensive income (FVOCI) securities, and gains or losses on cash flow hedging relationships. The attribution entry is as follows:

Dr Other comprehensive income to Non-controlling interests (statement of profit or loss and other comprehensive income)

Cr Non-controlling interests (statement of financial position)

Non-controlling interests are always presented on an *after-tax basis*. Hence, the attribution of profit to non-controlling interests in the income statement is shown below the profit after tax line. Any adjustment to non-controlling interests should be reflected on an after-tax basis (if tax adjustments are required).

The attribution of profit to non-controlling interests is not an expense item and should not be shown above the profit after tax line. The attribution is simply giving to non-controlling interests their share of profit. Without the attribution, the retained earnings of the group would be overstated and the non-controlling interests' share of equity would be understated. An example of the presentation of attribution of profit to non-controlling interests is shown in the annual report extracts of Keppel Corporation Limited below:

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### Annual report extract

## Consolidated Profit and Loss Account

For the financial year ended 31 December 2018

| | Note | 2018 $'000 | 2017 $'000 |
|---|---|---|---|
| **Revenue** | 23 | **5,964,781** | 5,963,773 |
| Materials and subcontract costs | | **(4,187,631)** | (3,957,402) |
| Staff costs | 24 | **(987,830)** | (1,027,019) |
| Depreciation and amortisation | | **(182,386)** | (212,380) |
| Impairment loss on financial assets | | **95,457** | (130,110) |
| Other operating income - net | | **531,089** | 164,184 |
| **Operating profit** | 25 | **1,042,566** | 801,046 |
| One-off financial penalty & related costs[i] | | **–** | (618,722) |
| Investment income | 26 | **9,991** | 19,871 |
| Interest income | 26 | **164,260** | 137,928 |
| Interest expenses | 26 | **(198,443)** | (189,227) |
| Share of results of associated companies | 9 | **221,518** | 290,533 |
| **Profit before tax** | | **1,239,892** | 441,429 |
| Taxation | 27 | **(283,747)** | (244,049) |
| **Profit for the year** | | **956,145** | 197,380 |
| Attributable to: | | | |
| **Shareholders of the Company** | | **943,829** | 196,025 |
| **Non-controlling interests** | 5 | **2,316** | 1,355 |
| | | **956,145** | 197,380 |
| Earnings per ordinary share | 28 | | |
| – basic | | **52.0 cts** | 10.8 cts |
| – diluted | | **51.7 cts** | 10.7 cts |

[i] One-off financial penalty and related costs arose from Keppel Offshore & Marine's global resolution with criminal authorities in the United States, Brazil and Singapore and related legal, accounting and forensics costs.



# Consolidated Statement of Comprehensive Income

For the financial year ended 31 December 2018

| | 2018 $'000 | 2017 $'000 |
|---|---|---|
| **Profit for the year** | **956,145** | 197,380 |
| Items that may be reclassified subsequently to profit and loss account: | | |
| Available-for-sale assets | | |
| – Fair value changes arising during the year | – | 1,619 |
| – Realised and transferred to profit and loss account | – | (28,815) |
| Cash flow hedges | | |
| – Fair value changes arising during the year, net of tax | **(238,794)** | 357,211 |
| – Realised and transferred to profit and loss account | **132,017** | (49,852) |
| Foreign exchange translation | | |
| – Exchange difference arising during the year | **(132,866)** | (220,787) |
| – Realised and transferred to profit and loss account | **5,574** | (9,537) |
| Share of other comprehensive income of associated companies | | |
| – Available-for-sale assets | – | 719 |
| – Cash flow hedges | **20,031** | (8,384) |
| – Foreign exchange translation | **(42,821)** | (93,232) |
| | **(256,859)** | (51,058) |
| Items that will not be reclassified subsequently to profit and loss account: | | |
| Financial assets, at FVOCI | | |
| – Fair value changes arising during the year | **(31,566)** | – |
| Foreign exchange translation | | |
| – Exchange difference arising during the year | **(3,545)** | (17,311) |
| Share of other comprehensive income of associated companies | | |
| – Financial assets, at FVOCI | **581** | – |
| | **(34,530)** | (17,311) |
| **Other comprehensive expense for the year, net of tax** | **(291,389)** | (68,369) |
| **Total comprehensive income for the year** | **664,756** | 129,011 |
| **Attributable to:** | | |
| Shareholders of the Company | **656,303** | 144,491 |
| Non-controlling interests | **8,453** | (15,480) |
| | **664,756** | 129,011 |

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### Amortization of Fair Value Differentials

Since non-controlling interests are credited with a share of the excess of fair value over book value (or debited with any shortfalls of fair value from book value) of identifiable assets and liabilities as at the date of acquisition, they must also bear the subsequent amortization of these fair value differentials. For example, if a fixed asset of an acquiree is undervalued at acquisition date, the higher depreciation charge subsequent to acquisition reduces the non-controlling shareholders' share of current profit after tax.

### Allocation of Dividends to Non-controlling Interests

Dividends declared by a subsidiary offset the dividend income of its parent. However, the offset will not be perfect if non-controlling interests are present. For example, Subsidiary A declares dividends of $1,600,000 to its shareholders. Parent owns 60% of Subsidiary A. In this situation, the allocation of dividends between parent and non-controlling interests, and the consolidation elimination entries are as follows:

$$\begin{aligned}\text{Dividends attributable to non-controlling interests} &= \text{Non-controlling interests \%} \times \text{Dividends declared by subsidiary} \\ &= 40\% \times \$1{,}600{,}000 \\ &= \$640{,}000\end{aligned}$$

$$\begin{aligned}\text{Dividend income}^{2} \text{ recorded by parent} &= \text{Parent's ownership interests \%} \times \text{Dividends declared by subsidiary} \\ &= 60\% \times \$1{,}600{,}000 \\ &= \$960{,}000\end{aligned}$$

The consolidation elimination entry is as follows:

| | | |
|---|---|---|
| Dr Dividend income (parent) . . . . . | 960,000 | |
| Dr Non-controlling interests (SFP) . . . . . | 640,000 | |
| Cr Dividends declared (subsidiary) . . . . . | | 1,600,000 |

The elimination entry reverses the profit and loss effects of dividends in the consolidated income statement. As the subsidiary is partially owned, dividends declared will not offset dividend income fully. The debit to non-controlling interests in the consolidated statement of financial position (SFP) indicates that non-controlling interests' residual stake in the net assets of the subsidiary has been reduced by the dividends declared. Dividends attributable to non-controlling interests are effectively a repayment of profits by a subsidiary. After elimination, dividends declared by a subsidiary should be a nil figure. The only dividends reported in the consolidated financial statements are those that are declared by the parent company.



### Analytical Check on the Non-controlling Interests' Balance

Essentially, non-controlling interests in a subsidiary have a share in the same components that the parent has under the acquisition method if the fair value basis is adopted. However, if non-controlling interests are recognized as a proportion of fair value of identifiable net assets, only two of the three components in Figure 4.4 apply to non-controlling interests. The two components are the share of book value of net assets or shareholders' equity of a subsidiary and the share of the balance of unamortized fair value adjustments. [If non-controlling interests have both present ownership interests (for example, ordinary shares) and potential ownership interests (for example, options), only present ownership interests may be measured as a proportion of identifiable net assets. The potential ownership interests have to be measured at full fair value.]

Since non-controlling interests have a share in shareholders' equity of a subsidiary, any "unrealized profit" included in the subsidiary's retained earnings have to be removed from the retained earnings before allocation to non-controlling interests. Unrealized profit from upstream sale (this is a sale from the subsidiary to the parent, which is further discussed in Chapter 5) is a profit that is not yet earned from the group's perspective, and should be adjusted out from the subsidiary's retained earnings. The profit is also deemed as unrealized from the non-controlling interests' point of view.

Hence, under the fair value basis, we can decompose non-controlling interests as shown in Figure 4.4.

**FIGURE 4.4 Components of non-controlling interests**

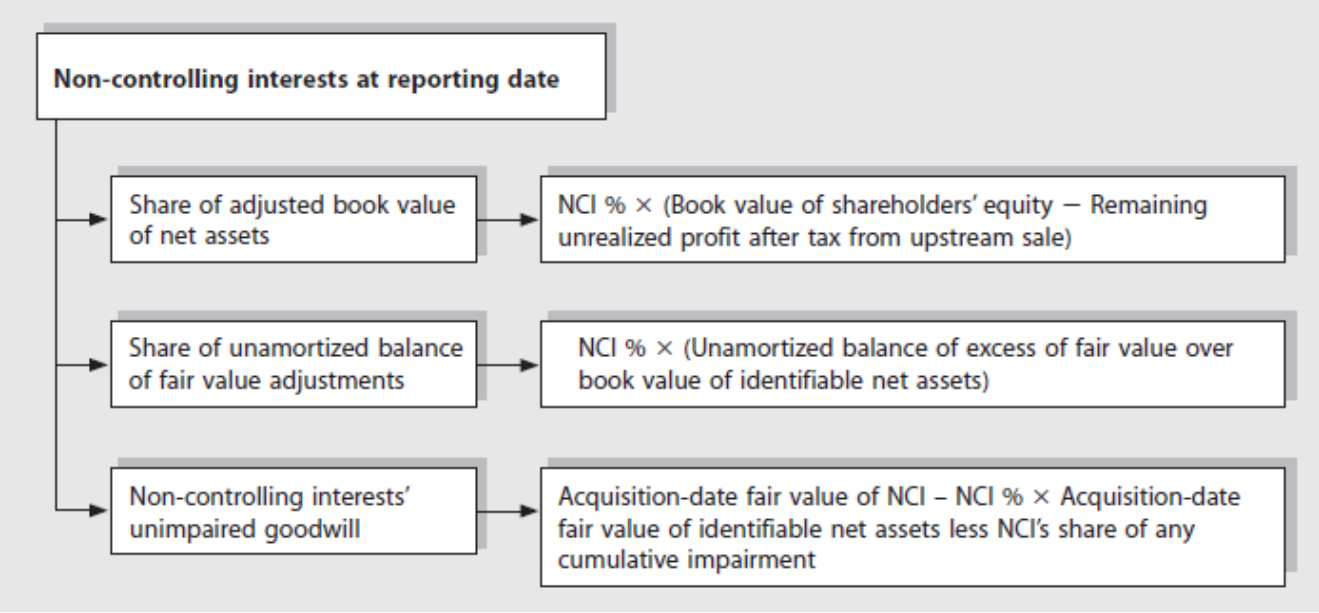





Table 4.1 illustrates the movement in non-controlling interests' balance.

**TABLE 4.1 Analytical check on non-controlling interests' balance**

| Event | Components | | |
|---|---|---|---|
| | Book value of identifiable net assets | Fair value less book value of identifiable net assets | Goodwill |
| Date of acquisition (re-enacted each year) | Share of book value of net assets or equity of subsidiary as at acquisition date | Share of fair value adjustments of identifiable net assets as at acquisition date | Non-controlling interests' goodwill |
| From acquisition date to beginning of current period (re-enacted each year) | +/– Share of post-acquisition profit / (loss) of subsidiary<br>+/– Share of other changes in equity | –/+ Past cumulative amortization of fair value adjustments of identifiable net assets | – Past cumulative impairment of goodwill |
| Current period | +/– Share of profit/loss of subsidiary after tax<br>– Share of dividends | –/+ Current amortization of fair value adjustments of identifiable net assets | – Current impairment of goodwill |

| | | | |
|---|---|---|---|
| | +/– Share of other changes in equity | | |
| Non-controlling interests' balance at year-end | = Share of adjusted book value of net assets or equity of subsidiary at end of current period | = Share of unamortized balance of fair value adjustments of identifiable net assets at year-end | = Non-controlling interests unimpaired goodwill |

Table 4.1 shows that under the fair value basis, non-controlling interests balances comprise the following three components:

1. Book value of equity,
2. Difference between fair value and book value of identifiable net assets; and
3. Goodwill.

For item (1), non-controlling interests have a share in both the upward and downward movements in changes in book value of equity (that is, the amount that is recognized in the legal entity financial statements of the subsidiary). For items 2 and 3, only extinguishment or impairment of fair value differentials and goodwill are recognized. Extinguishment or impairment of fair value differentials and goodwill will reduce the balance of non-controlling interests. They reflect that the initial net assets attributable to non-controlling interests have declined in carrying amount.

Hence, the balance of non-controlling interests as determined from the sum of consolidation journal entries can be checked with reference to the reporting date balance of the components above.

The equation as shown in Table 4.1 is as follows:

$$\text{Non-controlling interests' balance} = \left( \begin{array}{l} \text{Non-controlling interests' share of} \\ 1.\ \text{book value of net assets or equity of subsidiary,} \\ 2.\ \text{unamortized balance of fair value adjustment; and} \\ 3.\ \text{unimpaired balance of goodwill.} \end{array} \right)$$



The equation shows the relationship between non-controlling interests in the statement of financial position under IFRS 3 and the subsidiary's equity. The equation provides an indirect way of checking the non-controlling interests' balance in the consolidation worksheet. However, the check is not a fool-proof check on accuracy. If one of the components is in error, the check will affirm consistency but not the accuracy of the balance.

Illustrations 4.4 and 4.7 show how the check may be used. Several other illustrations in Chapters 5–7 demonstrate how the analytical check may be used in more complex settings.

### Can Non-controlling Interests Be a Debit Balance?

Non-controlling interests' share of losses in a subsidiary may exceed their share of the subsidiary's share capital, retained earnings, and other equity items. IFRS 10 paragraph B94 (Appendix B) requires the non-controlling interests to have a deficit balance under such a circumstance. This is a departure from the previous standard IAS 27 (revised 2003) that requires non-controlling interests to be carried at zero balance, with further losses being borne by the majority shareholders, unless the non-controlling interests have a binding obligation to make further investments to make good the losses, and they have the ability to do so.

There are two views on the merits of permitting non-controlling interests to be a debit balance. One opposing view is that the parent who has control of the subsidiary should bear the responsibility of supporting an insolvent subsidiary through guarantees or additional funding. Although a parent may not have any legal responsibility to support its

subsidiary in insolvency, it is seen by the business community to be giving an implicit guarantee through the preparation of consolidated financial statements (refer to Chapter 2 for the discussion on the underlying rationale for preparing consolidated financial statements). Proponents of this view also support the fact that the parent should absorb the negative equity of non-controlling interests as the latter have no control over the operations of the subsidiary. Using the limited liability argument, one can also say that non-controlling interests stand to lose only their investment in the subsidiary and have no legal or constructive obligation to bear any further losses.

On the other hand, the International Accounting Standards Board (the Board) concluded that non-controlling interests as equity holders participate proportionally in the risks and rewards of a subsidiary and that the parent, as with the non-controlling interests, has no further obligation to contribute assets to a failing subsidiary.[3] We show an example of how non-controlling interests can be a debit balance in Illustration 4.3. This is an extension of Illustration 4.2.

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### ILLUSTRATION 4.3 Non-controlling interests as a debit balance

Assume the facts as presented in Illustration 4.2. In 20x5, Subsidiary A had become insolvent. The intangible asset and goodwill that arose on acquisition date was fully impaired as at 31 December 20x5. The financial statements of Subsidiary A as at 31 December 20x5 are shown below. Show the balance of non-controlling interests in Subsidiary A as at 31 December 20x5.

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**Subsidiary A**
**Statement of Financial Position**
**As at 31 December 20x5 (current reporting date)**

| | |
|---|---|
| Net assets | $(100,000) |
| Equity: | |
| Share capital | $ 140,000 |
| Retained earnings | (240,000) |
| | $(100,000) |

Although the fair value of non-controlling interests as at acquisition date was a positive amount of $25,000, it had turned into a deficit balance of $10,000 as at current reporting date. Both parent and non-controlling interests will share proportionally in the losses. The parent of Subsidiary A will recognize its share of the negative equity of $90,000 in its consolidated statement of financial position. The consolidated statement of financial position will include an excess of liabilities over assets of $100,000 on the one side and negative non-controlling interests of $10,000 and a loss in retained earnings of $90,000 on the other side.

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## Comprehensive Example of Accounting for Non-Controlling Interests

Illustration 4.4 shows the re-enacted entries for non-controlling interests in consolidation using the fair value basis for non-controlling interests. The illustration also shows the analytical check of non-controlling interests that substantiates the balance of non-controlling interests to the three components as explained in Table 4.1.

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### ILLUSTRATION 4.4 Accounting for non-controlling interests

Pluto Co acquired a 60% interest in Saturn Co on 2 January 20x1 when Saturn's share capital and retained earnings were $80,000 and $30,000 respectively. Book values at the date of acquisition were close to fair values. Fair value of non-controlling interests as at the date of acquisition was $75,000. A control premium was paid by Pluto to acquire Saturn. The accounting policy is to recognize non-controlling interests at fair value at acquisition date. The following financial statements relate to Pluto Co and Saturn Co for the year ended 31 December 20x8.

**Income Statement and Partial Statement of Changes in Equity**
**For Year Ended 31 December 20x8**

| | Pluto Co | Saturn Co |
|---|---|---|
| Operating profit | $160,000 | $60,000 |
| Dividend income from Saturn | 18,900 | 0 |
| Net profit before tax | $178,900 | $60,000 |
| Tax expense | (48,900) | (18,000) |
| Net profit after tax | $130,000 | $42,000 |
| Retained earnings, 1 January | 110,000 | 38,200 |
| Dividends declared | (100,000) | (31,500) |
| Retained earnings, 31 December | $140,000 | $48,700 |



**Statement of Financial Position**
**As at 31 December 20x8**

| | Pluto Co | Saturn Co |
|---|---|---|
| Investment in Saturn Co cost | $117,000 | $ 0 |
| Other assets | 578,000 | 294,700 |
| *Total assets* | $695,000 | $294,700 |
| Share capital | $300,000 | $ 80,000 |
| Retained earnings | 140,000 | 48,700 |
| Long-term liabilities | 200,000 | 130,000 |
| Current liabilities | 55,000 | 36,000 |
| *Equity and liabilities* | $695,000 | $294,700 |

***Required:***

1. Show the consolidation journal entries that have to be passed for the year ended 31 December 20x8.
2. Perform an analytical check on the non-controlling interests as at 31 December 20x8.
3. Prepare the consolidation worksheets for the year ended 31 December 20x8.

(Illustration 4A.1 in Appendix 4A shows the answers for 1–3 above prepared under the alternative accounting policy of recognizing non-controlling interests as a proportion of identifiable net assets of subsidiary at acquisition date).

**1. Consolidation journal entries that have to be passed for the year ended 31 December 20x8**

*CJE1: Elimination of investment in Saturn as at date of acquisition*

| | | |
|---|---|---|
| Cr Non-controlling interests (SFP) . . . . . | | 16,800 |

Non-controlling interests' share of current profit = 40% × $42,000 = $16,800

*CJE4: Elimination of dividends declared by Saturn*

| | | |
|---|---|---|
| Dr Dividend income (Pluto) (60% × $31,500) . . . . . | 18,900 | |
| Dr Non-controlling interests (SFP) (40% × $31,500) . . . | 12,600 | |
| Cr Dividends declared by Saturn . . . . . | | 31,500 |

***Explanatory notes:***

*CJE1:* This journal entry has to be re-enacted each year for as long as Pluto has an investment in Saturn. Recall that consolidation is a combination of the separate legal entity financial statements of Pluto and Saturn. Since the separate financial statements of parent include the investment account, there will be a double-counting of assets if the investment account is added with the individual assets and liabilities of the subsidiary. Non-controlling interests are measured at fair value at acquisition date in accordance with the stated accounting policy. Hence, recognized goodwill of $82,000 includes non-controlling interests' goodwill as at acquisition date.

*CJE2:* Since CJE1 sets up the non-controlling interests' position as at acquisition date, the share of changes in post-acquisition retained earnings (that is, the difference between the retained earnings from the date of acquisition to the beginning of the year) has to be credited to the non-controlling interests' account. Dividends paid during the post-acquisition period up to 1 January 20x8 would be included in the net change figure. The combined effect of CJE1 and CJE2 reinstates the balance of non-controlling interests as at the beginning of the current period.

A similar entry would be passed if Saturn had cumulative other comprehensive income as at the beginning of the current period. Another consolidation journal entry would allocate share of change in post-acquisition other comprehensive income to non-controlling interests.



*CJE3:* This entry essentially reduces consolidated profit after tax retained by Pluto's shareholders by allocating non-controlling interests' share to the non-controlling interests' account in the statement of financial position. If a subsidiary is profitable, the entry to the income statement will be a debit entry (that is, an allocation of profit to non-controlling interests); if the subsidiary is loss-making, a credit entry will indicate that part of the loss is borne by the non-controlling interests.

A similar entry would be passed if Saturn had current other comprehensive income. Non-controlling interests would have a share of current other comprehensive income.

*CJE4:* Non-controlling interests' share of dividends is debited to the non-controlling interests' account in the statement of financial position to show a partial recovery or realization of the cumulative profits attributable to non-controlling interests.

**2. Analytical check on the non-controlling interests as at 31 December 20x8**

Reconstruct the non-controlling interests account in the statement of financial position:

| Non-controlling interests | | | |
|---|---|---|---|
| Dividends as repayment of profit (CJE4) ... | $12,600 | As at date of acquisition (CJE1) ..... | $75,000 |
| | | Share of change in retained earnings to beginning of year (CJE2) ..... | 3280 |
| Balance at 31 December 20x8 ..... | 82,480 | Share of current profit after tax (CJE3) ... | 16,800 |
| | $95,080 | | $95,080 |

*Analytical check on non-controlling interests:*

| | | |
|---|---|---|
| Saturn's book value of net assets as at 31 December 20x8 | $128,700 | |
| Non-controlling interests' share of net assets . . . . . | $ 51,480 | (40% × $128,700) |
| Non-controlling interests' goodwill . . . . . | 31,000 | (refer to CJE1) |
| Non-controlling interests' balance . . . . . | $ 82,480 | |



### 3. Consolidation worksheets for the year ended 31 December 20x8

| Income Statement and Partial Statement of Changes in Equity For Year Ended 31 December 20x8 | | | | | |
|---|---|---|---|---|---|
| | | | Consolidation adjustments | | |
| | Pluto Co | Saturn Co | Dr | Cr | Consolidated totals |
| Operating profit ..... | $ 160,000 | $ 60,000 | | | $220,000 |
| Dividend income from Saturn ..... | 18,900 | 0 | $ 18,900 | | 0 |
| Net profit before tax..... | $ 178,900 | $ 60,000 | | | $220,000 |
| Tax expense ..... | (48,900) | (18,000) | | | (66,900) |
| Net profit after tax..... | $ 130,000 | $ 42,000 | | | $153,100 |
| Profit attributable to non-controlling interests ..... | | | 16,800 | | (16,800) |
| Profit attributable to shareholders ..... | $130,000 | $ 42,000 | | | $136,300 |
| Retained earnings, 1 January ..... | 110,000 | 38,200 | 30,000<br>3,280 | | 114,920 |
| Dividends declared ..... | (100,000) | (31,500) | | 31,500 | (100,000) |
| Retained earnings, 31 December ..... | $ 140,000 | $ 48,700 | $ 68,980 | $ 31,500 | $151,220 |

| Statement of Financial Position<br>As at 31 December 20x8 | | | | | |
|---|---|---|---|---|---|
| | | | Consolidation adjustments | | |
| | Pluto Co | Saturn Co | Dr | Cr | Consolidated totals |
| Investment in Saturn, cost | $117,000 | $ 0 | | $117,000 | $ 0 |
| Other assets | 578,000 | 294,700 | | | 872,700 |
| Goodwill | | | $ 82,000 | | 82,000 |
| Total assets | $695,000 | $294,700 | $ 82,000 | $117,000 | $954,700 |
| Share capital | $300,000 | $ 80,000 | $ 80,000 | | $300,000 |
| Retained earnings | 140,000 | 48,700 | 68,980 | $ 31,500 | 151,220 |
| Non-controlling interests | | | 12,600 | 75,000 | 82,480 |
| | | | | 3,280 | |
| | | | | 16,800 | |
| Long-term liabilities | 200,000 | 130,000 | | | 330,000 |
| Current liabilities | 55,000 | 36,000 | | | 91,000 |
| Total liabilities and equity | $695,000 | $294,700 | $161,580 | $126,580 | $954,700 |
| | | | $243,580 | $243,580 | |

***Explanatory notes:***

- The building blocks of the worksheet are the separate financial statements of Pluto and Saturn.
- A debit and credit column is included for consolidation adjustments.
- The consolidated totals are from the summation of the separate financial statement amounts and the consolidation adjustments.
- The total of debit and credit adjustments from the income statement will be transferred to retained earnings in the statement of financial position.
- The total of debit and credit adjustments from the income statement will not be equal; however, the total of debit and credit adjustments must be equal in the statement of financial position.

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## ILLUSTRATION 4.5 Presentation of Non-controlling Interests

Consider the presentation of non-controlling interests in the consolidated financial statements of Pluto and Saturn from Illustration 4.4:

| Consolidated Income Statement<br>For Year Ended 31 December 20x8 | |
|---|---|
| Net profit before tax | $220,000 |
| Tax expense | (66,900) |
| Net profit after tax | $153,100 |
| Profit attributable to non-controlling interests | $ 16,800 |
| Profit attributable to parent's shareholders | 136,300 |
| | $153,100 |

**Consolidated Statement of Changes in Equity**
**For Year Ended 31 December 20x8 (extract)**

| | |
|---|---|
| Retained earnings, 1 January | $114,920 |
| Net profit after tax attributable to parent's shareholders | 136,300 |
| Dividends declared | (100,000) |
| Retained earnings, 31 December | $151,220 |

**Statement of Financial Position**
**As at 31 December 20x8 (condensed)**

| | |
|---|---|
| Other assets | $872,700 |
| Goodwill | 82,000 |
| Total assets | $954,700 |
| Share capital | $300,000 |
| Retained earnings | 151,220 |
| Parent's shareholders' equity | $451,220 |
| Non-controlling interests | 82,480 |
| Total equity | $533,700 |
| Long-term liabilities | $330,000 |
| Current liabilities | 91,000 |
| Total liabilities | $421,000 |
| Total liabilities and equity | $954,700 |



Here are a few noteworthy points with respect to the presentation of non-controlling interests:

1. Consolidated profit after tax of $153,100 is the combined profit of Pluto and Saturn (after eliminating intercompany transactions) that has to be allocated to Pluto's shareholders and the non-controlling interests of Saturn.
2. Non-controlling interests are presented as a separate component of equity in the statement of financial position. Non-controlling interests in the statement of financial position represent the residual interests that the non-controlling shareholders of a partially-owned subsidiary have in the net assets of that subsidiary.
3. Since non-controlling interests are a separate component of equity, the terms "retained earnings" and "share capital" in the consolidated statement of financial position belong wholly to the parent's shareholders.

Illustration 4.6 shows the amortization entries that are required when non-controlling interests are measured at fair value on acquisition date. The amortization entries under the alternative basis when non-controlling interests are measured as a proportion of identifiable net assets are the same. Illustration 4A.2 in Appendix 4A shows the entries prepared under the accounting policy where non-controlling interests have a proportionate share of identifiable net assets at acquisition date.

---

## ILLUSTRATION 4.6 Amortization of fair value differentials

P Co paid a cash consideration of $6,200,000 and issued 1,000,000 of its own shares to acquire 80% ownership interest in S Co on 1 January 20x5. The fair value of P Co's shares at the date of acquisition was $3 per share. The book and fair values of assets, liability and contingent liability, and the remaining useful lives of leased property and in-process research and development (R&D) of S Co as at date of acquisition are as follows:

| | Book value | Fair value | Remaining useful life |
|---|---|---|---|
| Leased property . . . | $4,000,000 | $5,000,000 | 20 years |
| In-process R&D . . . | | 2,000,000 | 10 years |
| Other assets . . . . . | 1,900,000 | 1,900,000 | |
| Liabilities . . . . . | (1,200,000) | (1,200,000) | |
| Contingent liability . | | (100,000) | |
| Net assets . . . . . | $4,700,000 | $7,600,000 | |
| Share capital . . . . . | $1,000,000 | | |
| Retained earnings . | 3,700,000 | | |
| Shareholders' equity | $4,700,000 | | |

***Additional information:***

(a) The contingent liability of $100,000 met the recognition criteria in IFRS 3. The contingent liability was recognized as a provision for loss by the acquiree in its legal entity financial statements in December 20x5. The differences between fair value and book value of identifiable net assets above gave rise to a deferred tax liability at a tax rate of 20%.

(b) Fair value of non-controlling interests as at the date of acquisition was $2,300,000, which was proportional to the fair value of consideration transferred by P Co (that is, P Co did not pay a control premium or enjoyed a discount on the acquisition). The fair value of non-controlling interests was recognized as at the date of acquisition.



(c) Net profit after tax of S Co for the year ended 31 December 20x5 was $1,000,000. No dividends were declared during 20x5. Shareholders' equity as at 31 December 20x5 was $5,700,000.

***Required:***

1. Prepare the consolidation adjustments for P Co for 20x5.
2. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x5.

**Determination of consideration transferred**

$$\begin{aligned}\text{Consideration transferred} &= \text{Cash consideration} + \text{Fair value of shares issued} \\ &= \$6{,}200{,}000 + (1{,}000{,}000 \times \$3) \\ &= \$6{,}200{,}000 + \$3{,}000{,}000 \\ &= \$9{,}200{,}000\end{aligned}$$

The consideration transferred comprises the acquirer's interest in the following:

- Book value of identifiable net assets
- Difference between fair value over book value of identifiable net assets:
  - Excess of fair value over book value of the leased property;
  - Fair value of in-process R&D (since this is identifiable and has a reliably measurable fair value, it should be recognized in the business combination);

- Fair value of contingent liability (although not recognized by the acquiree, it meets the criteria in IFRS 3 and has a fair value that is reliably measurable and should be recognized in the business combination); and
- Deferred tax liability (recognized on the fair value adjustments).

- Goodwill (the residual)

**1. Consolidation adjustments for P Co in 20x5**

*CJE1: Elimination of investment in subsidiary as at acquisition date*

| | | | | |
|---|---|---|---|---|
| Dr Share capital . . . . . | 1,000,000 | | | |
| Dr Opening retained earnings | 3,700,000 | | | |
| Dr Leased property . . . . . | 1,000,000 | | | |
| Dr In-process R&D . . . . . | 2,000,000 | | | |
| Dr Goodwill . . . . . | 4,480,000 | (Note 2) | | |
| Cr Contingent liability . . | | | 100,000 | |
| Cr Deferred tax liability (net) | | | 580,000 | (Note 1) |
| Cr Investment in S Co . . | | | 9,200,000 | |
| Cr Non-controlling interests | | | 2,300,000 | (Note 2) |



**Note 1:**

$$\text{Deferred tax liability on fair value adjustments} = \text{Tax rate} \times \left( \begin{array}{c} \text{Excess of fair value} \\ \text{over book value of} \\ \text{leased property} \end{array} + \begin{array}{c} \text{Fair value of} \\ \text{in-process} \\ \text{R\&D} \end{array} - \begin{array}{c} \text{Fair value of} \\ \text{contingent} \\ \text{liability} \end{array} \right)$$

$$= 20\% \times (\$1{,}000{,}000 + \$2{,}000{,}000 - \$100{,}000)$$
$$= 20\% \times \$2{,}900{,}000$$
$$= \$580{,}000$$

***Explanatory note:***

The net increase in assets arising from the fair value adjustments of $2,900,000 implies higher future taxable income. Hence, future tax payable on the future taxable income has to be recognized in the form of a deferred tax liability. The deferred tax asset arising from the contingent liability is offset against the deferred tax liabilities. IAS 12 *Income Taxes* requires an entity to offset deferred tax assets and deferred tax liabilities if there is a legal right of set-off of current tax assets and current tax liabilities; and the related income taxes are levied by the same taxation authority on the same taxable entity or different taxable entities that intend to settle on a net basis (paragraph 74).

**Note 2:**

$$\text{Goodwill} = \text{Consideration transferred} + \text{Amount of non-controlling interests at acquisition date} - \text{Recognized amount of identifiable net assets at acquisition date}$$

$$= \$9{,}200{,}000 + \$2{,}300{,}000 - (\$7{,}600{,}000 - \$580{,}000)$$
$$= \$11{,}500{,}000 - \$7{,}020{,}000$$
$$= \$4{,}480{,}000$$

Since non-controlling interests' fair value is proportional to the fair value of consideration transferred by P Co, the goodwill attributable to non-controlling interests is 20% of entity goodwill of $4,480,000.

*CJE2: Depreciation and amortization of excess of fair value over book value for the year ended 31 December 20x5*

| | | | | |
|---|---|---|---|---|
| Dr Depreciation of leased property | 50,000 | ($1 million/20 years) | | |
| Dr Amortization of in-process R&D | 200,000 | ($2 million/10 years) | | |
| Cr Accumulated depreciation | | | | 50,000 |
| Cr Accumulated amortization | | | | 200,000 |

*CJE3: Reversal of entry relating to provision for loss*

| | | | | |
|---|---|---|---|---|
| Dr Provision for loss | | | 100,000 | |
| Cr Loss expense | | | | 100,000 |

***Explanatory note:***

Since a contingent liability was already recognized on 1 January 20x5 in the consolidated financial statements, the recognition of a provision by the acquiree in its legal entity financial statements results in double-counting; hence, this reversal entry is necessary. The contingent liability expense should not be recognized as a post-acquisition expense. The effect of the reversal entry is to increase net earnings of the acquiree by $100,000.



*CJE4: Tax effects on amortization of fair value adjustments and entry relating to provision for lawsuit*

| | | | | |
|---|---|---|---|---|
| Dr Deferred tax liability (net) | | | 30,000 | |
| Cr Tax expense | | | | 30,000 |

Tax effects are as follows:

| | |
|---|---|
| Depreciation of excess of fair value over book value of leased property | $ 50,000 |
| Amortization of in-process R&D | 200,000 |
| Reversal of loss from contingent liability | (100,000) |
| Subsequent amortization of fair value adjustments | $150,000 |
| Tax rate | 20% |

| | |
|---|---|
| Reduction in tax expense arising from amortization . . . . . | $ 30,000 |

***Explanatory note:***

Since the amortization of fair value adjustments results in a reduction in profit of $150,000, a consequent reduction in tax expense of $30,000 arises. The reduction in tax expense is a reversal of the deferred tax liability that was recognized in CJE1.

| *CJE5: Allocation of current income to non-controlling interests* | | |
|---|---|---|
| Dr Income to non-controlling interests . . . . . | 176,000 | |
| Cr Non-controlling interests . . . . . | | 176,000 |
| Net profit after tax . . . . . | | $1,000,000 |
| Excess depreciation . . . . . | | (50,000) |
| Excess amortization . . . . . | | (200,000) |
| Reversal of loss from contingent liability . . . . . | | 100,000 |
| Tax effects on fair value adjustments . . . . . | | 30,000 |
| Adjusted net profit after tax . . . . . | | $ 880,000 |
| Non-controlling interests' share at 20% . . . . . | | $ 176,000 |

***Explanatory note:***

CJE5 allocates income to non-controlling interests. As explained in Table 4.1, non-controlling interests have a share in the extinguishment of the initial fair value differences and in the impairment of goodwill. (If non-controlling interests are recognized as a proportion of identifiable net assets, they will share in only the extinguishment of the initial fair value differences and not the impairment of goodwill). Note that the top line item in CJE5's working is the net profit after tax that represents the increase in the book value of equity of the subsidiary. The other adjustments relate to the extinguishment of the fair value differentials. Non-controlling interests fall in carrying amount when the initial net assets carried at fair value are used, consumed, sold, or depreciated. Overall, non-controlling interests have a share of $176,000 of adjusted profit, which represents the increase in book value and a decrease in the fair value differentials that were initially allocated to non-controlling interests.



**2. Analytical check on the balance of non-controlling interests as at 31 December 20x5**

First, reconstruct the balance of non-controlling interests as at 31 December 20x5.

| | |
|---|---|
| Non-controlling interests as at acquisition date (CJE1) . . . . . | $2,300,000 |
| Income allocated to NCI for 20x5 (CJE5) . . . . . | 176,000 |
| Non-controlling interests as at 31 December 20x5 . . . . . | $2,476,000 |

Next, reconcile the balance to the three components that non-controlling interests have a share in.

| | | |
|---|---|---|
| Book value of identifiable net assets as at 31 December 20x5 | $ 5,700,000 | |
| Unamortized balance of fair value adjustments as at 31 December 20x5: | | |
| Leased property ($1,000,000 × 19/20) . . . . . | | 950,000 |

| | | |
|---|---|---|
| In-process R&D ($2,000,000 × 9/10) | 1,800,000 | |
| | $ 2,750,000 | |
| After-tax unamortized balance at 80% . . . . . | 2,200,000 | |
| Adjusted net assets of S Co . . . . . | $7,900,000 | |
| Non-controlling interests at 20% . . . . . | $ 1,580,000 | (ALT 2) |
| Goodwill attributable to non-controlling interests . . | 896,000 | |
| | $ 2,476,000 | (ALT 1) |

***Explanatory notes:***

Goodwill attributable to non-controlling interests is included in the balance of non-controlling interests as we are recognizing non-controlling interests at full fair value on acquisition date (we describe this as Alternative 1). If non-controlling interests are recognized as a proportion of identifiable net assets at acquisition date, goodwill will not be included in the final balance of non-controlling interests (we describe this as Alternative 2). The calculation of goodwill attributable to non-controlling interests is shown in the shaded box below. In this illustration, goodwill attributable to non-controlling interests is proportional to goodwill of the acquirer because there is no control premium.

- Control premium is determined by comparing the consideration transferred by the parent and the fair value of non-controlling interests. (It may appear circular, but essentially, the fair value of non-controlling interests should be determined independently, for example, through valuation techniques). If the consideration transferred is proportionally more than the fair value of non-controlling interests, there is a control premium. In the opposite situation, a control discount (which often arises in a fire sale) arises. In this example, consideration transferred is $9.2 million and the fair value of non-controlling interests is $2.3 million. The ratio of the two is 4:1, which is consistent with the ratio of their ownership interests.
- In other illustrations where the fair value of non-controlling interests is not proportional to fair value of consideration transferred by the parent (for example, where the parent pays a control premium to acquire a subsidiary), the goodwill attributable to non-controlling interests must be separately identified in the reconciliation.
- We show here how the goodwill for non-controlling interests (NCI) is calculated:

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Goodwill attributable to non-controlling interests
= Acquisition-date fair value of non-controlling interests — NCI's share of acquisition-date fair value of identifiable net assets less NCI's share of any cumulative impairment of goodwill
= $2,300,000 – 20% × $7,020,000
= $2,300,000 – $1,404,000
= $896,000

---

Illustration 4.7 explains the consolidation adjustments that are passed in two consecutive years if non-controlling interests are measured at fair value. Illustration 4A.3 in Appendix 4A shows the equivalent entries if the non-controlling interests are measured as the proportion of fair value of identifiable net assets, excluding goodwill.

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## ILLUSTRATION 4.7 Multi-year consolidation

On 31 December 20x1, P Co purchased 90% of S Co's outstanding ordinary shares from S Co's present shareholders for $8.26 million cash and issuance of one million shares of P Co's ordinary shares, which had a market value of $5 million. The statements of financial position of S Co and P Co at the date of acquisition (also the financial year-end)

are shown below. Assume that all fair value adjustments have tax effects and give rise to either a deferred tax liability or deferred tax asset. However, goodwill impairment loss does not have any tax effects. Fair value of non-controlling interests at acquisition date was $1,400,000. Tax rate was 20% throughout. Assume an accounting policy where the non-controlling interests are recognized at fair value on acquisition date.

| | S Co | | P Co |
|---|---|---|---|
| | Book value | Fair value | Book value |
| Assets: | | | |
| Land | $ 780,000 | $ 1,170,000 | $ 1,560,000 |
| Leased buildings – net | 5,200,000 | 6,500,000 | 10,400,000 |
| Equipment – net | 2,600,000 | 2,210,000 | 9,100,000 |
| Investment in S Co | | | 13,260,000 |
| Inventories | 650,000 | 780,000 | 1,170,000 |
| Receivables – net | 390,000 | 390,000 | 910,000 |
| Other current assets | 520,000 | 520,000 | 780,000 |
| Cash | 260,000 | 260,000 | 1,690,000 |
| | $10,400,000 | $11,830,000 | $38,870,000 |
| Equity and liabilities: | | | |
| Ordinary shares | $ 6,500,000 | | $25,870,000 |
| Retained earnings | 1,170,000 | $ 8,970,000* | 5,590,000 |
| Other liabilities | 910,000 | 910,000 | 2,600,000 |
| Loan payable | 1,820,000 | 1,820,000 | 4,810,000 |
| Contingent liabilities | | 130,000 | |
| | $10,400,000 | $11,830,000 | $38,870,000 |

* Fair value of identifiable net assets as at the date of acquisition.



Information on S Co's net profit after tax and dividends declared during 20x2 are as follows:

| | |
|---|---|
| P Co's dividend income from S | $409,500 |
| S Co's net profit after tax | 728,000 |
| S Co's dividends declared | 455,000 |

There were no other changes to equity. In 20x2, the following information applies to undervalued and overvalued assets and goodwill:

(a) Undervalued inventories of $130,000 – sold in 20x2

(b) Undervalued land of $390,000 – still held by S Co; no depreciation

(c) Undervalued buildings of $1,300,000 – useful life 50 years from 1 January 20x2

(d) Overvalued equipment[4] of $390,000 – useful life five years from 1 January 20x2

(e) Contingent liabilities of $130,000 – materialized (paid off) in 20x2 (these liabilities met the recognition criteria of IFRS 3)

(f) Goodwill — impairment loss of $520,000 in 20x2 recognized

In 20x3, the following information applies:

(g) Undervalued land of $390,000 — land was sold in 20x3

(h) Undervalued buildings of $1,300,000 — useful life 50 years from 1 January 20x2

(i) Overvalued equipment of $390,000 — useful life five years from 1 January 20x2

(j) Net profit after tax of S Co for 20x3 — $400,000

(k) No dividends were declared by S Co during 20x3.

(l) Recognize the tax effects of fair value adjustments.

Deferred tax assets should be set-off against deferred tax liabilities of S Co.

***Required:***

1. Prepare the consolidation adjustments for 20x2.
2. Prepare the consolidation adjustments for 20x3.

**Schedule of unamortized excess of fair value over book value**

| Unamortized excess of fair value over book value | Adjustment in group's statement of financial position on 31 December 20x1 | Amortization of excess in group's income statement in 20x2 | Impact on non-controlling interest's share of profit after tax |
|---|---|---|---|
| Undervalued inventories | Inventories increased by $130,000 | Net profit reduced by $130,000 (higher cost of sales) | NCI's share of profit reduced by $13,000 |
| Undervalued land | Land increased by $390,000 | No impact because land is not depreciable | No impact |
| Undervalued buildings | Buildings increased by $1,300,000 | Net profit reduced by $26,000 (higher depreciation of buildings) | NCI's share of profit reduced by $2600 |
| Overvalued equipment | Equipment decreased by $390,000 | Net profit increased by $78,000 (lower depreciation of equipment) | NCI's share of profit increased by $7,800 |
| Contingent liabilities | Contingent liabilities increased by $130,000 | Net profit increased by $130,000 (contingent expense reduced by $130,000). Refer to explanatory note. | NCI's share of profit increased by $13,000 |
| Deferred tax liability (net) | Deferred tax liability (net) increased by $260,000, that is, 20% of fair value adjustments above ($130,000 + $390,000 + $1,300,000 – $390,000 – $130,000) | Net increase to net profit before tax from the above adjustments = $52,000<br>Tax expense on the increased profit = 20% × $52,000 = $10,400<br>Refer to the explanatory notes below on the rationale for the deferred tax entry. | NCI's share of tax on increased profit = 10% × $10,400 = $1,040 |
| Goodwill | Goodwill increased by $5,950,000 Refer to computation | Impairment loss of $520,000 is recognized as an expense | NCI has a share of goodwill impairment loss of $52,000 |

**Computation of goodwill**

| | |
|---|---|
| Consideration transferred | |
| Cash | $ 8,260,000 |
| Value of new P Co shares issued | 5,000,000 |
| | $13,260,000 |
| Fair value of non-controlling interests at acquisition date | 1,400,000 |
| | $14,660,000 |
| Less net fair value of identifiable net assets (Note 1) | (8,710,000) |
| Goodwill (Note 2) | $ 5,950,000 |



**Note 1:**

| | |
|---|---|
| *Fair value of net assets of S Co* | |
| Fair value of assets | $11,830,000 |
| Less fair value of liabilities | (2,730,000) |
| Contingent liabilities | (130,000) |
| | $ 8,970,000 |
| Fair value adjustments: | |
| Fair value of identifiable net assets | $ 8,970,000 |
| Book value of identifiable net assets | (7,670,000) |
| Excess | $ 1,300,000 |
| Deferred tax liability (net) on fair value adjustments | $ (260,000) |
| After-tax fair value of identifiable net assets | $ 8,710,000 |

**Note 2:**

$$\text{Goodwill attributable to parent} = \$13{,}260{,}000 - (90\% \times \$8{,}710{,}000)$$
$$= \$5{,}421{,}000$$

$$\text{Goodwill attributable to non-controlling interests} = \$1{,}400{,}000 - (10\% \times \$8{,}710{,}000)$$
$$= \$529{,}000$$

In this illustration, goodwill attributable to parent and non-controlling interests are not proportional to each other because of a control premium paid by the parent.

**1. Consolidation adjustments for 20x2**

| | |
|---|---|
| *CJE1: Elimination of investment in S Co as at acquisition date* | |
| Dr Ordinary shares (S Co) | 6,500,000 |

| | | |
|---|---|---|
| Dr Opening retained earnings (S Co) | 1,170,000 | |
| Dr Goodwill | 5,950,000 | |
| Dr Inventories | 130,000 | |
| Dr Land | 390,000 | |
| Dr Buildings – net | 1,300,000 | |
| Cr Investment in S Co | | 13,260,000 |
| Cr Equipment – net | | 390,000 |
| Cr Contingent liabilities | | 130,000 |
| Cr Deferred tax liability (net) | | 260,000 |
| Cr Non-controlling interests | | 1,400,000 |



*CJE2: Impairment of goodwill and amortization of excess of fair values of identifiable net assets recognized in the income statement (I/S) and statement of financial position (SFP)*

| | | |
|---|---|---|
| Dr Cost of sales (I/S) | 130,000 | |
| Dr Depreciation – buildings (I/S) | 26,000 | |
| Dr Impairment of goodwill (I/S) | 520,000 | |
| Dr Accumulated depreciation – equipment (*SFP*) | 78,000 | |
| Dr Contingent liabilities (*SFP*) | 130,000 | |
| Dr Tax expense (I/S) | 10,400 | |
| Cr Inventories (*SFP*) | | 130,000 |
| Cr Accumulated depreciation – buildings (*SFP*) | | 26,000 |
| Cr Goodwill (*SFP*) | | 520,000 |
| Cr Depreciation – equipment (I/S) | | 78,000 |
| Cr Contingent liability expense (I/S) | | 130,000 |
| Cr Deferred tax liability (net) (*SFP*) | | 10,400 |

***Explanatory note:***

What is the rationale for the consolidation adjustment on contingent liabilities? Remember that the contingent liabilities were accrued in the consolidated statement of financial position as at 31 December 20x1, but not in the individual statement of financial position of S Co as at that date. On payment, in 20x2, S Co recorded the following entries in its financial statements:

| | | | |
|---|---|---|---|
| (a) | Dr Expense | 130,000 | |
| | Cr Cash | | 130,000 |

However, the group had accrued for such a liability on 31 December 20x1 in the consolidated financial statements. On payment, the entry at the group level should have been:

| | | | |
|---|---|---|---|
| (b) | Dr Contingent liabilities | 130,000 | |
| | Cr Cash | | 130,000 |

Since the consolidation worksheets are built upon the separate financial statements of a parent and its subsidiaries, we need to put through a consolidation entry to adjust the effects of entry (a) above to be in line with entry (b).

Hence, the consolidation adjustment reverses out the debit entry to expense and shows the reduction of the contingent liability.

| | | | |
|---|---|---|---|
| (c) | Dr Contingent liabilities | 130,000 | |
| | Cr Expense | | 130,000 |

***Explanatory note:***

What is the rationale for the deferred tax entry? A deferred tax liability was set up as at date of acquisition to recognize the potential increase in future tax payable that will arise when the fair value adjustments are realized. These fair value adjustments are realized when the asset is sold or depreciated.

In the financial statements of S Co, depreciation and cost of sale expense are determined based on book values and not their fair values. Hence, relative to the group's perspective, S's net profit before tax for Year 2 is lower by $52,000 than what it should be.



| | Increase/Decrease in pre-tax profit of S |
|---|---|
| *As recorded in S's books:* | |
| Lower cost of sales relating to undervalued inventory | $ 130,000 |
| Lower depreciation charge on buildings | 26,000 |
| Higher depreciation charge on equipment | (78,000) |
| Increase in expense on contingency | (130,000) |
| Net decrease in net profit before tax | $ (52,000) |

Thus, the tax expense recorded by S Co based on book values is lower by $10,400 (20% of $52,000) than is the case if fair values at 31 December 20x1 had been used as a basis to amortize and depreciate the assets. Since the tax expense of S Co is understated from the group's perspective, a consolidation adjustment has to be passed:

| | | |
|---|---|---|
| Dr Tax expense | 10,400 | |
| Cr Deferred tax liability (net) | | 10,400 |

The tax expense entry in 20x2 increases the deferred tax liability because the amortization adjustments are income-increasing in 20x2. A reversal of the deferred tax liability arises only from 20x3 onward when the amortization adjustments become income-decreasing. The implicit deferred tax assets are set-off against the deferred tax liabilities since both relate to the same taxable entity and the same taxation authority.

***Explanatory note:***

Why are the cost of sale, depreciation, and other adjustments to the fair value adjustments not multiplied by the parent's ownership interest? The consolidated net profit after tax relates to the entity as a whole, and is presented before the allocation of the subsidiary's after-tax profit to non-controlling interests. Consolidation entries that show the subsequent amortization of undervalued or overvalued assets or liabilities are adjusted against "net profit before tax"; the related deferred tax adjustments are adjusted against "tax expense." Since consolidated "net profit before tax" and "tax expense" relate to the combined shares of both controlling and non-controlling interests, the

consolidation adjustments to these lines should reflect the total adjustments and not the parent's share of the adjustments. However, after the allocation of profit after tax to non-controlling interests, the profit attributable to shareholders belongs purely to the shareholders of the parent. This final figure enters into the retained earnings' balance. Retained earnings as shown in the consolidated financial statements are attributable wholly to the parent company's shareholders, but consolidated net profit after tax is attributable to both the parent company's shareholders and the non-controlling interests. Hence, any consolidation adjustments to retained earnings must reflect only the parent's share of the adjustments, unlike consolidation adjustments to income statement items.

*CJE3: Allocation of non-controlling's share of subsidiary's profits for the year*

| | | |
|---|---|---|
| Dr Income to non-controlling interests (I/S) | 24,960 | |
| Cr Non-controlling interests (SFP) | | 24,960 |



***Workings:***

| | |
|---|---|
| Net profit after tax of S for 20x2 | $728,000 |
| Increase in cost of sale of undervalued inventories | (130,000) |
| Increase in depreciation on undervalued building | (26,000) |
| Decrease in depreciation on overvalued equipment | 78,000 |
| Decrease in expense on earlier recognized contingency | 130,000 |
| Increase in tax expense as a result of the above adjustments | (10,400) |
| Impairment loss on goodwill | (520,000) |
| Adjusted net profit after tax of S | $249,600 |
| NCI's share of adjusted net profit after tax at 10% | $ 24,960 |

***Explanatory note:***

Since the set-up entry recognizes non-controlling interests at fair value, non-controlling interests have a share of entity goodwill and, hence, are exposed to impairment loss on goodwill. IAS 36 *Impairment of Assets* requires the impairment loss to be pro-rated between the parent and non-controlling interests on the same basis as that on which profit or loss is allocated (IAS 36 App C: C6). In other words, the impairment loss is not pro-rated in accordance with the proportion of goodwill recognized by parent and non-controlling interests.

*CJE4: Elimination of dividends*

| | | |
|---|---|---|
| Dr Dividend income (I/S) | 409,500 | |
| Dr Non-controlling interests (SFP) | 45,500 | |
| Cr Dividends declared (I/S) | | 455,000 |

**Reconciliation of non-controlling interests with net assets of subsidiary**

First, reconstruct the non-controlling interests' balance from the consolidation adjustments.

| | | |
|---|---|---|
| Non-controlling interests at date of acquisition | $1,400,000 | (CJE1) |
| Non-controlling interests' share of current profit after tax | 24,960 | (CJE3) |
| Less dividends declared to non-controlling interests | (45,500) | (CJE4) |
| Non-controlling interests as at 31 December 20x2 | $1,379,460 | |

Next, determine the identifiable net assets or shareholders' equity of S Co and the implicit goodwill attributable to non-controlling interests. Non-controlling interests' balance in the statement of financial position should be equal to the sum of non-controlling interests' share of adjusted identifiable net assets or shareholders' equity of the subsidiary and non-controlling interests' goodwill.



| | |
|---|---|
| Book value of shareholders' equity as at 31 December 20x2 (Note 1) | $ 7,943,000 |
| Unamortized excess of fair value over book value at 31 December 20x2 (Note 2) | 1,081,600 |
| *Adjusted identifiable net assets of S Co* | $ 9,024,600 |
| Non-controlling interests' share of identifiable net assets | $ 902,460 |
| Non-controlling interests' unimpaired goodwill: | |
| Goodwill attributable to non-controlling interests | 529,000 |
| Impairment loss | (52,000) |
| | 477,000 |
| Non-controlling interests as at 31 December 20x2 | $ 1,379,460 |

**Note 1:** Book value of shareholders' equity as at 31 December 20x2

| | | |
|---|---|---|
| Share capital | | $6,500,000 |
| Retained earnings: | | |
| As at 1 January 20x2 | $1,170,000 | |
| Net profit after tax | 728,000 | |
| Dividends declared | (455,000) | |
| As at 31 December 20x2 | | 1,443,000 |
| Book value of shareholders' equity | | $7,943,000 |

**Note 2:** Unamortized excess of fair value over book value at 31 December 20x2

| | | |
|---|---|---|
| Undervalued land | $ 390,000 | |
| Undervalued buildings | 1,274,000 | ($1,300,000 × 49/50) |
| Overvalued equipment | (312,000) | ($390,000 × 4/5) |
| | $1,352,000 | |
| After-tax balance | $1,081,600 | ($1,352,000 × 80%) |

*Undervalued inventories and contingent liabilities have zero unamortized balances as at 31 December 20x2*

### 2. Consolidation adjustments for 20x3

**Schedule of unamortized excess of fair value over book value**

| Unamortized excess of fair value over | Adjustment to be re-enacted | Consolidation adjustment in 20x3 | Impact on non-controlling interests |
|---|---|---|---|

book value

| | | | |
|---|---|---|---|
| Undervalued inventories | Inventories increased by $130,000 | Since inventory was sold in 20x2 (cost of sales increased in 20x2), group retained earnings at beginning 20x3 decreased by (90% × $130,000) or $117,000 | NCI's share of opening retained earnings decreased by $13,000 |
| Undervalued land | Land increased by $390,000 | Profit on sale of land decreased by $390,000 because of the higher carrying amount of the land | NCI's share of profit decreased by $39,000 |
| Undervalued buildings | Buildings increased by $1,300,000 | Depreciation of building for current year increased by $26,000; opening retained earnings reduced by (90% × $26,000) or $23,400 to reflect the group's share of prior year's depreciation adjustment of $26,000 | NCI's share of profit reduced by $2,600; NCI's share of opening retained earnings reduced by $2,600 |
| Overvalued equipment | Equipment decreased by $390,000 | Depreciation of equipment decreased by $78,000; opening retained earnings increased by (90% × $78,000) or $70,200, which is the group's share of the previous year's adjustment to depreciation | NCI's share of profit increased by $7,800; NCI's share of opening retained earnings increased by $7,800 |
| Contingent liabilities | Contingent liabilities increased by $130,000 | Previous year's expense decreased by $130,000. Opening retained earnings increased by (90% × $130,000) or $117,000 | NCI's share of opening retained earnings increased by $13,000 |
| Deferred tax liability (net) | Deferred tax liability (net) increased by $260,000 | Lower tax expense arose as a result of the above adjustments<br>= 20% × ($390,000 + $26,000 – $78,000)<br>= $67,600<br>With respect to previous year's effects on deferred tax liability, opening retained earnings decreased by (90% × $10,400) or $9,360. | Because NCI's profit is reduced by $33,800 above, the tax expense would also fall. Hence, NCI's share of profit after tax would increase by 20% × $33,800 = $6,760<br>Previous year's effects would decrease non-controlling interests by $1,040 |
| Goodwill | Goodwill increased by $5,950,000 | Impairment loss of $520,000 in the previous year would reduce opening retained earnings by $468,000 (90% × $520,000) | NCI's share of opening retained earnings reduced by $52,000 |



*CJE1: Re-enactment of the elimination of investment in S Co at acquisition date*

| | | |
|---|---|---|
| Dr Ordinary shares (S Co) | 6,500,000 | |
| Dr Opening retained earnings (S Co) | 1,170,000 | |
| Dr Goodwill | 5,950,000 | |
| Dr Inventories | 130,000 | |
| Dr Land | 390,000 | |
| Dr Buildings – net | 1,300,000 | |
| Cr Investment in S Co | | 13,260,000 |
| Cr Equipment – net | | 390,000 |
| Cr Contingent liabilities | | 130,000 |
| Cr Deferred tax liability (net) | | 260,000 |
| Cr Non-controlling interests | | 1,400,000 |



CJE2: Goodwill impairment losses in 20x2

| | | |
|---|---|---|
| Dr Opening retained earnings | 468,000 | |
| Dr Non-controlling interests | 52,000 | |
| Cr Goodwill | | 520,000 |

***Explanatory note:***

Goodwill and goodwill impairment are not shown in the separate financial statements; hence, a consolidation adjustment is necessary to re-enact the entries. As non-controlling interests are measured at fair value at acquisition date, they have to bear a proportion of impairment losses. As goodwill impairment was debited to consolidated profit and loss in the previous year, we have to debit the opening retained earnings and non-controlling interests in the statement of financial position to reflect the prior-year entry.

Why is an adjustment to the "opening" retained earnings required? In the statement of changes in equity, the opening balance of retained earnings has to be adjusted to arrive at the same balance that was reported in the previous consolidated financial statements. (The word "opening" does not need to be inserted, but it is included here to differentiate between past and current profits.)

*CJE3: Allocation of post-acquisition retained earnings of S Co to non-controlling interests from date of acquisition to beginning of current period*

| | | |
|---|---|---|
| Dr Opening retained earnings | 27,300 | |
| Cr Non-controlling interests | | 27,300 |
| Retained earnings at beginning of current period (1 January 20x3) | | $1,443,000* |
| Retained earnings at acquisition date (31 December 20x1) | | 1,170,000 |
| Change in retained earnings | | $ 273,000 |
| Non-controlling interests' share at 10% | | $ 27,300 |

* Reconstructed, since there were no changes in retained earnings other than profit and dividends.

*CJE4: Prior-year disposal and depreciation of undervalued and overvalued*

| | | |
|---|---|---|
| *net assets* | | |
| Dr Accumulated depreciation – equipment (SFP) . . . . | 78,000 | |
| Dr Contingent liabilities (SFP) . . . . . | 130,000 | |
| Cr Inventories (SFP) . . . . . | | 130,000 |
| Cr Accumulated depreciation – buildings (SFP) . . . | | 26,000 |
| Cr Deferred tax liability (net) (SFP) . . . . . | | 10,400 |
| Cr Opening retained earnings (Note 1) . . . . . | | 37,440 |
| Cr Non-controlling interests (SFP) (Note 1) . . . . . | | 4,160 |

**Note 1:** Prior-year consolidation adjustments

| | Debit/Credit |
|---|---|
| Cost of sales . . . . . | $(130,000) |
| Depreciation of buildings . . . . . | (26,000) |
| Depreciation of equipment . . . . . | 78,000 |
| Realization of contingent liability . . . . . | 130,000 |
| Increase in tax expense (net) . . . . . | (10,400) |
| Total prior-year consolidation adjustments excluding goodwill impairment | $ 41,600 |



Adjustment to consolidated opening retained earnings (90% × $41,600) = $37,440
Allocated to non-controlling interests (10% × $41,600) = $4,160

***Explanatory note:***

We multiply the total prior-year consolidation adjustments by the parent's ownership interest to allocate to consolidated retained earnings because the latter (unlike consolidated profit after tax) belongs solely to the parent's shareholders. Non-controlling interests' share of the subsidiary's retained earnings is included separately in the non-controlling interests' balance. Any adjustment to consolidated retained earnings must relate to the parent's share only as retained earnings in the consolidated statement of financial position belong to the parent's shareholders and not the non-controlling interests.

As with the entry on goodwill impairment, this adjustment is necessary to show that the previous year's consolidated profit and loss were adjusted for the disposal, depreciation, and realization of the fair value adjustments of identifiable assets and the contingent liabilities. Past income effects that need to be re-enacted in consolidation are effected to opening retained earnings. In this adjustment, non-controlling interests have a share of the prior-year consolidation adjustments. Since non-controlling interests are credited with the fair value adjustments at acquisition date, the amortization of the fair value adjustments are debited to non-controlling interests subsequently. Table 4.1 presents the movements that affect non-controlling interests.

| | | |
|---|---|---|
| *CJE5: Current adjustments to undervalued and overvalued assets and liabilities* | | |
| Dr Profit on sale of land (I/S) . . . . . | 390,000 | |
| Dr Depreciation – Buildings (I/S) . . . . . | 26,000 | |
| Dr Accumulated depreciation – Equipment (SFP) . . . . | 78,000 | |

| | | |
|---|---|---|
| Dr Deferred tax liability (net) (SFP) . . . . . | 67,600 | |
| Cr Land (SFP) . . . . . | | 390,000 |
| Cr Accumulated depreciation – Buildings (SFP) . . . | | 26,000 |
| Cr Depreciation – Equipment (I/S) . . . . . | | 78,000 |
| Cr Tax expense (I/S) . . . . . | | 67,600 |

*CJE6: Non-controlling interests' share of subsidiary's profits for the year*

| | | |
|---|---|---|
| Dr Income to non-controlling interests (I/S) . . . . . | 12,960 | |
| Cr Non-controlling interests (SFP) . . . . . | | 12,960 |

***Workings:***

| | |
|---|---|
| Net profit after tax of S Co for 20x3 . . . . . | $400,000 |
| Current adjustments: | |
| Higher cost of sales of undervalued land . . . . . | (390,000) |
| Depreciation of undervalued buildings . . . . . | (26,000) |
| Depreciation of overvalued equipment . . . . . | 78,000 |
| Related tax on above . . . . . | 67,600 |
| Adjusted net profit after tax of S Co . . . . . | $129,600 |
| NCI's share of adjusted net profit after tax at 10% . . . . . | $ 12,960 |



**Reconciliation of non-controlling interests with net assets of subsidiary**

*Non-controlling interests' balance reconstructed from the consolidation adjustments:*

| | | | |
|---|---|---|---|
| Non-controlling interests at date of acquisition . . . . . | $1,400,000 | (CJE1) | |
| Non-controlling interests' share of past goodwill impairment . | (52,000) | (CJE2) | |
| Non-controlling interests' share of post-acquisition retained earnings to beginning of the current year | 27,300 | (CJE3) | |
| Non-controlling interests' share of prior years' amortization of fair value adjustments | 4,160 | (CJE4) | |
| Non-controlling interests' share of current profit after tax . . . | 12,960 | (CJE6) | |
| Non-controlling interests at 31 December 20x3 . . . . . | $1,392,420 | | |
| Book value of shareholders' equity as at 31 December 20x3 (Note 1) | | | $8,343,000 |
| Unamortized excess of fair value over book value at 31 December 20x3 (Note 2) | | | 811,200 |
| Adjusted identifiable net assets of S Co . . . . . | | | $9,154,200 |
| Non-controlling interests' share of adjusted identifiable net assets | | | $ 915,420 |
| Non-controlling interests' goodwill . . . . . | | | 477,000 |

| | | |
|---|---|---|
| Non-controlling interests as at 31 December 20x3 . . . . . | | $1,392,420 |

**Note 1:** Book value of shareholders' equity as at 31 December 20x3

| | | |
|---|---|---|
| Share capital . . . . . | | $6,500,000 |
| Retained earnings: | | |
| As at 1 January 20x3 . . . . . | $1,443,000 | |
| Net profit after tax . . . . . | 400,000 | |
| Dividends declared . . . . . | 0 | |
| As at 31 December 20x3 . . . . . | | 1,843,000 |
| Book value of shareholders' equity . . . . . | | $8,343,000 |

**Note 2:** Unamortized excess of fair value over book value at 31 December 20x3

| | | |
|---|---|---|
| Undervalued buildings . . . . . | $1,248,000 | ($1,300,000 × 48/50) |
| Overvalued equipment . . . . . | (234,000) | ($390,000 × 3/5) |
| | $1,014,000 | |
| After-tax balance . . . . . | $ 811,200 | ($1,014,000 × 80%) |

*Undervalued inventories and land and contingent liabilities have zero unamortized balances as at 31 December 20x3*



# Other Components of Non-Controlling Interests

It is not unusual for the acquiree to have equity components other than equity shares that are not owned directly or indirectly by the parent. These equity components may not be present ownership interests that entitle their holders to a proportionate share of the net assets of the entity in the event of liquidation. An example of this is the equity component of a compound financial instrument. Conversely, other components could represent present ownership interests but do not entitle their holders to proportionate share of net assets of the company during liquidation. This could take the form of preference shares for which the holders are entitled only to the amount they had paid on acquisition of the preference shares on liquidation.

Such equity components will fall within the definition of non-controlling interests as IFRS 3 defines non-controlling interests as "the equity in a subsidiary not attributable directly or indirectly to a parent." Hence, they will have to be considered as part of non-controlling interests in the formula discussed above to calculate the goodwill on acquisition.

Paragraph 19 of IFRS 3 stipulates the accounting treatment. In essence, all other components of non-controlling interests that are not present ownership interests and do not entitle their holders to a pro rata share of the entity's net assets in the event of liquidation are measured at fair value at the date of acquisition. The reason why they are measured at fair value (the option of measurement at the proportionate share of the acquiree's identifiable net assets is not extended to such instruments) is because these interests could potentially have been measured at zero balance under Alternative 2 as they do not entitle their holders to the entity's net assets. Consequently, third party economic interests will not be reflected in the acquisition accounting under Alternative 2, and this is contrary to the principles

underpinning IFRS 3. The following table shows a list of examples of non-controlling interests and their corresponding measurement basis.

| Acquiree's equity components | Measurement basis under IFRS 3 |
|---|---|
| Ordinary shares (held by NCI) | Proportion of identifiable net assets or at fair value |
| Preference shares with holders entitled to proportionate share of net assets upon liquidation | Proportion of net identifiable assets at fair value |
| Preference shares with holders **not** entitled to proportionate share of net assets upon liquidation | Fair value |
| Equity element of compound financial instruments | Fair value |
| Options/Warrants | Fair value |
| Employee share options[5] | Market based measure in accordance with IFRS 2 at acquisition date |

Essentially, what the measurement requirement in IFRS 3 paragraph 19 means is that the component of non-controlling interest in the formula to compute goodwill comprises two components, that is, non-controlling interests that are present ownership interests, which entitle their holders to proportionate share in the identifiable net assets and those that are not. Both will have to be taken into consideration in calculating goodwill.

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| | | | | | |
|---|---|---|---|---|---|
| Goodwill = | Fair value of consideration transferred | + Amount of non-controlling interests at acquisition date | | + Fair value of acquirer's previously held interest in the acquire | − Recognized net identifiable assets of acquiree measured in accordance with IFRS 3 |
| | = Fair value of consideration transferred | + NCI that are present ownership interests and entitle holders to a pro rata share of net assets | + NCI that are **not** present ownership interests and **do not** entitle holders to a pro rata share of net assets | + Fair value of acquirer's previously held interest in the acquire | − Recognized net identifiable assets of acquiree measured in accordance with IFRS 3 |
| | **(Fair value)** | **(Choice of proportionate share of identifiable assets or fair value)** | **(Fair value)** | **(Fair value)** | **(Generally fair value)** |

Appendix 4B presents a comprehensive illustration to show the accounting effect of the above.

## GOODWILL IMPAIRMENT TESTS

IAS 36 *Impairment of Assets* requires goodwill to be reviewed annually for impairment loss. Because goodwill is an integral asset in a business unit, it is not assessed for impairment as a stand-alone asset, but is reviewed for impairment as part of a cash-generating unit. In a "bottom-up" approach, goodwill has to be allocated to each of the acquirer's cash-generating units, or groups of cash-generating units, which are expected to benefit from the synergies of the business combination (IAS 36:80). A cash-generating unit is:

1. The lowest level at which the goodwill is monitored for internal management purposes; and
2. Is not larger than a segment determined under IFRS 8 *Operating Segments.*

If the "carrying amount" of a cash-generating unit is greater than its "recoverable amount," an impairment loss arises. The "carrying amount" is the book value of the net assets of the cash-generating unit. The "recoverable amount" is defined as the higher of fair value less costs to sell off the cash-generating unit and its "value in use." Value in use is the present value of future net cash flows arising from the cash-generating unit.

Hence, the first step is to determine the "carrying amount" of a cash-generating unit. Effectively, each cash-generating unit should have its own statement of financial position, and a "carrying amount" should be determined for the net assets of that cash-generating unit. Because the "carrying amount" is compared with the "recoverable amount" of a cash-generating unit as a whole, the carrying amount must include allocated goodwill that is attributable to both the parent and the non-controlling interests.

As discussed earlier in the chapter, IFRS 3 allows two bases of measurement of non-controlling interests as at acquisition date, that is, either at fair value or as a proportion of the fair value of identifiable net assets of the acquiree. Table 4.2 shows the impact of the basis of measurement on goodwill impairment tests.

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**TABLE 4.2 Basis of measurement of non-controlling interests and goodwill impairment**

| Item | Non-controlling interests at fair value at acquisition date | Non-controlling interests as a proportion of identifiable net assets at acquisition date |
|---|---|---|
| Goodwill in consolidated financial statements | Includes non-controlling interests' goodwill | Does not include non-controlling interests' goodwill |
| Carrying amount of cash-generating unit | Goodwill is allocated to cash-generating unit without further adjustment | Goodwill has to be grossed-up to include non-controlling interests' share<br>Notionally adjusted goodwill = Recognized goodwill/Parent's interest |
| Impairment loss | Impairment loss is shared between parent and non-controlling interests on same basis on which profit or loss is allocated | Impairment loss is borne only by parent as goodwill for non-controlling interests is not recognized |

The next step is to determine the "recoverable amount" of the cash-generating unit. Fair value less costs to sell requires an arm's length measure of the price that would be received to sell the cash-generating unit in an orderly transaction between market participants at the measurement date (refer to IFRS 13 *Fair Value Measurement* for elaboration of the definition). Fair value less costs to sell requires use of market-based inputs or market participants' assumptions in the valuation process. Value in use, on the other hand, uses internal or entity-specific inputs to determine the future cash flows that can be generated from a cash-generating unit. Value in use is likely to be more discretionary as assumptions about future cash flows are required. IAS 36, in an anti-conservative stance, allows the higher of the two metrics to be used to determine "recoverable amount." While fair value less costs to sell may be the more objective metric, it may be impracticable at times to determine this measure in the absence of arm's length

benchmarks. Hence, IAS 36 allows an alternative metric in the form of the value in use to be applied in the goodwill impairment test.

If the carrying amount of a cash-generating unit is greater than its recoverable amount, IAS 36 requires the impairment loss to be first allocated to goodwill, and the balance to other assets in proportion to their individual carrying amounts. Goodwill impairment tests have to be carried out on an annual basis, regardless of whether indications of impairment exist. An impairment loss on goodwill, once made, cannot be reversed if subsequent information reveals that the impairment no longer exists. This prohibition does not apply to most other assets; however, fair value through other comprehensive income securities of an equity nature is also subject to this prohibition. The stringency of this rule is probably due to the difficulty of disentangling the effects of a reversal of past loss events from the creation of new internally generated goodwill. Since IAS 38 *Intangible Assets* prohibits the recognition of internally generated goodwill in the absence of an acquisition event, reversing a goodwill impairment loss may inadvertently result in the recognition of new internally generated goodwill.



Figure 4.5 illustrates the decision process on goodwill impairment.

**FIGURE 4.5 Goodwill impairment tests**

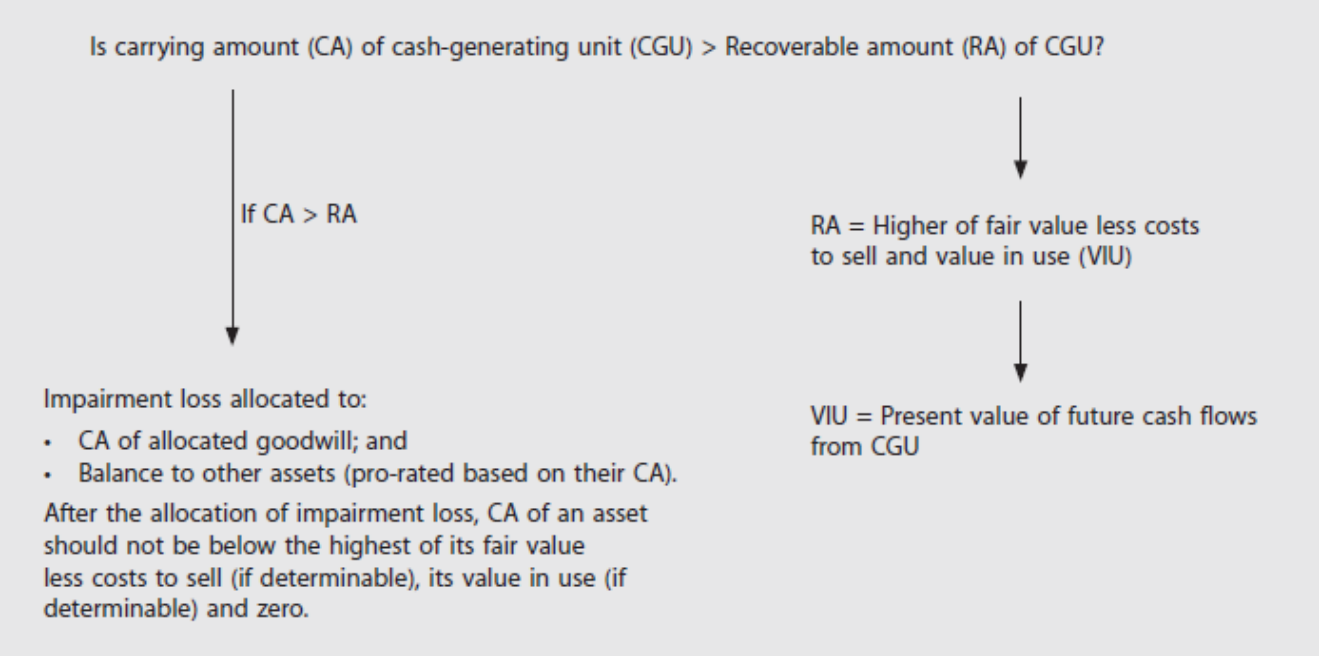



Illustration 4.8 demonstrates the principles of the goodwill impairment tests.

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**ILLUSTRATION 4.8 Goodwill impairment tests**

Company X has 80% ownership in a cash-generating unit (CGU) that had identifiable net assets of $6 million as at 31 December 20x1. The recoverable amount of the CGU as an entity was $5 million as at that date. Determine the impairment loss of the goodwill in the CGU under the following two alternative measurement bases:

(a) Non-controlling interests measured at fair value at acquisition date. Goodwill recognized by the cash-generating unit was $1,200,000.

(b) Non-controlling interests measured as a proportion of fair value of identifiable net assets at acquisition date. Goodwill recognized by the cash-generating unit was $1,000,000.

**Alternative (a): Non-controlling interests at acquisition-date fair value**

| | Goodwill | Identifiable net assets | Total |
|---|---|---|---|
| Carrying amount . . . . . | $1,200,000 | 6,000,000 | $7,200,000 |
| Recoverable amount . . . . . | | | 5,000,000 |
| Impairment loss . . . . . | 1,200,000 | 1,000,000 | $2,200,000 |
| Impairment loss borne by parent and non-controlling interests | 1,200,000 | 1,000,000 | $2,200,000 |



***Explanatory notes:***

- Goodwill has to be allocated to a cash-generating unit to enable a comparison between the carrying amount of all assets of the unit (including goodwill) and the recoverable amount.
- Under alternative (a), goodwill attributable to non-controlling interests is included under recognized goodwill. Hence, no further adjustment is required to goodwill.
- Since carrying amount is greater than recoverable amount by $2,200,000, the impairment loss is deemed to arise from goodwill first; in this example, the entire goodwill amount of $1,200,000 is deemed impaired and the remaining $1,000,000 of impairment loss is deemed to arise from the identifiable net assets.

**Alternative (b): Non-controlling interests as a proportion of identifiable net assets**

| | Goodwill | | Identifiable net assets | Total |
|---|---|---|---|---|
| Carrying amount . . . . . | $1,000,000 | | 6,000,000 | $7,000,000 |
| Non-controlling interests' unrecognized share of goodwill | 250,000 | (20% × $1 million/0.8) | | 250,000 |
| Notionally adjusted carrying amount | $1,250,000 | | 6,000,000 | $7,250,000 |
| Recoverable amount . . . . . | | | | 5,000,000 |
| Impairment loss . . . . . | $1,250,000 | | 1,000,000 | $2,250,000 |
| Impairment loss recognized . . | 1,000,000 | (80% × $1.25 million) | 1,000,000 | $2,000,000 |

***Explanatory note:***

Since comparison is done against the carrying amount of assets of a CGU, goodwill is regrossed, under alternative (b), to show the theoretical goodwill as at the date of acquisition. By regrossing recognized goodwill, non-controlling interests' unrecognized share of goodwill is included in the analysis.

---

In Illustration 4.8, the notionally adjusted carrying amount exceeds the recoverable amount of the cash-generating unit. The resulting impairment loss is deemed to arise from the theoretical goodwill first, and the balance of the loss is then pro-rated among the identifiable net assets of the cash-generating unit in proportion to their carrying amounts. Since the theoretical goodwill is shared by both parent and non-controlling shareholders, the impairment loss is pro-

rated to show the effective impairment loss suffered by the parent company. The amount of $1,000,000, which is the parent's share of total impairment loss, is debited to the consolidated income statement.

## CONCLUSION

Two sets of financial statements must be presented in many countries: the investor's separate financial statements and the consolidated financial statements for the group of companies. A "group" is defined in IFRS 10 as comprising a parent and its subsidiaries. As economic boundaries extend through "control" over the financial and operating policies of subsidiaries, another level of reporting is required to show the performance, financial position, and cash flows of the economic entity.

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Although two sets of accounts have to be prepared, only one set of "books" or "ledgers" has to be kept by the legal entity. Instead, consolidation worksheets are prepared at the end of each reporting period to prepare consolidated financial statements. The "consolidation equation" is a summation of the line items of the financial statements of the parent and its subsidiaries, and an incorporation of adjustments to eliminate and adjust intragroup transactions and balances. Hence, consolidation elimination entries and adjustments have to be understood with reference to the original entries that are passed in the individual books of either the parent or the subsidiary. Finally, the transactions and balances in the consolidated financial statements must reflect the group's, and not the legal entity's, perspective.

Consolidation entails an "asset substitution" process whereby the investment in a subsidiary is replaced with identifiable assets and liabilities of the subsidiary and a residual asset called goodwill. The "asset substitution" process is governed by IFRS 3, which requires the use of the acquisition method to allocate the consideration transferred. Under the acquisition method, the acquirer is deemed to have obtained control of all the assets and liabilities of the acquiree. The acquisition date is a critical economic event that signifies an exchange of economic resources between the acquirer and the former owners. Hence, the use of fair values is applied to recognize the assets and liabilities as at the date of acquisition. Further, unrecognized intangible assets and contingent liabilities are also recognized if they meet the recognition criteria in IFRS 3. The requirements in IFRS 3 indicate a shift toward the entity theory of consolidation whereby a consistent treatment is applied to reflect the interests of both controlling and non-controlling shareholders. Non-controlling interests are also included as a component in equity. The methodology of applying the specific principles of IFRS 3 with respect to the acquisition method and accounting for non-controlling interests in multiple periods are illustrated in this chapter.

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APPENDIX 4A

# Illustrations of Non-controlling Interests Measured as a Proportion of Acquisition-date Identifiable Net Assets

## ILLUSTRATION 4A.1

Refer to the question in Illustration 4.4. If we assume that the accounting policy of the acquirer is to recognize non-controlling interests as a proportion of the recognized amount of the identifiable net assets of the acquiree, the answers to the question in Illustration 4.4 will be as follows:

**1. Consolidation journal entries that have to be passed for the year ended 31 December 20x8**

| | | |
|---|---|---|
| *CJE1: Elimination of investment in Saturn as at date of acquisition* | | |
| Dr Share capital (Saturn) | 80,000 | |
| Dr Opening retained earnings (Saturn) | 30,000 | |
| Dr Goodwill | 51,000 | |
| Cr Investment in Saturn | | 117,000 |
| Cr Non-controlling interests | | 44,000 |

Under the alternative accounting policy, the amount of non-controlling interests is measured as a proportion of the recognized amount of identifiable net assets at acquisition date. In this example, the book value of $110,000 (the shareholders' equity at acquisition date) approximates fair value of identifiable net assets at acquisition date.

Hence, non-controlling interests = 40% × $110,000 = $44,000

$$\text{Goodwill} = \left(\begin{matrix}\text{Consideration}\\ \text{transferred}\end{matrix} + \begin{matrix}\text{Amount of}\\ \text{non-controlling interests}\end{matrix}\right) - \begin{matrix}\text{Recognized amount of identifiable}\\ \text{net assets at acquisition date}\end{matrix}$$

$$= \$117{,}000 + \$44{,}000 - \$110{,}000$$

$$= \$51{,}000$$

Other consolidation adjusting entries in Illustration 4.4 apply and are not repeated here.



**2. Analytical check on the non-controlling interests as at 31 December 20x8**

Reconstruct the non-controlling interests account in the statement of financial position:

**Non-controlling interests**

| | | | |
|---|---|---|---|
| Dividends as repayment of profit (CJE4) ... | $12,600 | As at date of acquisition (CJE1) ..... | $44,000 |
| | | Share of change in retained earnings to beginning of year (CJE2) ..... | 3,280 |
| Balance at 31 December 20x8 ..... | 51,480 | Share of current profit after tax (CJE3) ..... | 16,800 |
| | $64,080 | | $64,080 |

| | |
|---|---|
| Analytical check on non-controlling interests: | |
| Saturn's book value of net assets as at 31 December 20x8 | $128,700 |
| Non-controlling interests' share of net assets at 40% | 51,480 |

Under the alternative accounting policy, non-controlling interests do not have a share of the entity's goodwill.

## ILLUSTRATION 4A.2

Refer to the question in Illustration 4.6. If we assume that the accounting policy of the acquirer is to recognize non-controlling interests as a proportion of the recognized amount of the identifiable net assets of the acquiree, the answers to the question in Illustration 4.6 will be as follows:

**1. Consolidated adjustments for 20x5**

*CJE1: Elimination of investment in subsidiary as at acquisition date*

| | | | | |
|---|---|---|---|---|
| Dr Share capital . . . . . | 1,000,000 | | | |
| Dr Opening retained earnings | 3,700,000 | | | |
| Dr Leased property . . . . . | 1,000,000 | | | |
| Dr In-process R&D . . . . . | 2,000,000 | | | |
| Dr Goodwill . . . . . | 3,584,000 | (Note 3) | | |
| Cr Contingent liabilities . | | | 100,000 | |
| Cr Deferred tax liability (net) | | | 580,000 | (Note 1) |
| Cr Investment in S Co . . | | | 9,200,000 | |
| Cr Non-controlling interests | | | 1,404,000 | (Note 2) |

**Note 1:**

$$\text{Deferred tax liability on fair value adjustments} = \text{Tax rate} \times \left( \text{Excess of fair value over book value of leased property} + \text{Fair value of in-process R\&D} - \text{Fair value of contingent liabilities} \right)$$

$$= 20\% \times (\$1{,}000{,}000 + \$2{,}000{,}000 - \$100{,}000)$$
$$= 20\% \times \$2{,}900{,}000$$
$$= \$580{,}000$$



**Note 2:** Non-controlling interests measured as a proportion of identifiable net assets at acquisition date

| | |
|---|---|
| Fair value of identifiable assets and liabilities . . . . . | $7,600,000 |
| Deferred tax liability on fair value adjustments . . . . . | (580,000) |
| | $7,020,000 |
| Non-controlling interests' share at 20% . . . . . | $1,404,000 |

**Note 3:**

$$\text{Goodwill} = \left(\begin{matrix}\text{Consideration}\\ \text{transferred}\end{matrix} + \begin{matrix}\text{Amount of non-controlling}\\ \text{interests at acquisition date}\end{matrix}\right) - \begin{matrix}\text{Recognized amount of identifiable}\\ \text{net assets at acquisition date}\end{matrix}$$

$$= \$9{,}200{,}000 + \$1{,}404{,}000 - (\$7{,}600{,}000 - \$580{,}000)$$
$$= \$10{,}604{,}000 - \$7{,}020{,}000$$
$$= \$3{,}584{,}000$$

Other consolidation adjusting entries in Illustration 4.6 apply and are not repeated here.

**2. Analytical check on non-controlling interests as at 31 December 20x5**

First, reconstruct the balance of non-controlling interests as at 31 December 20x5.

| | |
|---|---|
| Non-controlling interests as at acquisition date | $1,404,000 |
| Income allocated to NCI for 20x5 | 176,000 |
| Non-controlling interests as at 31 December 20x5 | $1,580,000 |

Next, reconcile the balance to the two components (excluding goodwill) that non-controlling interests have a share in.

| | | |
|---|---|---|
| Book value of identifiable net assets as at 31 December 20x5 | | $5,700,000 |
| After-tax unamortized balance of fair value adjustments as at 31 December 20x5: | | |
| Leased property ($1,000,000 × 19/20) | 950,000 | |
| In-process R&D ($2,000,000 × 9/10) | 1,800,000 | |
| | $2,750,000 | |
| After-tax balance at 80% | | 2,200,000 |
| Adjusted net assets of S Co | | $7,900,000 |
| Non-controlling interests at 20% | | $1,580,000 |

---

## ILLUSTRATION 4A.3

Refer to the question in Illustration 4.7. The solutions below show the consolidation adjustments and analytical check on non-controlling interests assuming the accounting policy of measuring non-controlling interests as a proportion of identifiable net assets at acquisition date. Explanatory notes are included in Illustration 4.7 with the exception of explanations that apply to this accounting policy. Impairment loss recognized on consolidation is $468,000, which relates to the parent's share only.



**Computation of goodwill**

| | |
|---|---|
| Consideration transferred: | |
| Cash | $ 8,260,000 |
| Value of new P Co shares issued | 5,000,000 |
| | $13,260,000 |

| | |
|---|---|
| Amount of non-controlling interests at acquisition date (Note 2) | 871,000 |
| | $14,131,000 |
| Less net fair value of identifiable net assets (Note 1) | (8,710,000) |
| Goodwill attributable to parent's share | $ 5,421,000 |

**Note 1:**

| | |
|---|---|
| *Fair value of net assets of S Co:* | |
| Fair value of assets | $11,830,000 |
| Less fair value of liabilities | (2,730,000) |
| Contingent liabilities | (130,000) |
| | $ 8,970,000 |
| *Fair value adjustments:* | |
| Fair value of identifiable net asset | $ 8,970,000 |
| Book value of identifiable net assets | (7,670,000) |
| Excess | $ 1,300,000 |
| Deferred tax liability (net) on fair value adjustments | $ (260,000) |
| After-tax fair value of identifiable net assets | $ 8,710,000 |

**Note 2:**

Non-controlling interests as a proportion of after-tax fair value of identifiable net assets
= 10% × $8,710,000
= $871,000

**1. Consolidation adjustments for 20x2**

*CJE1: Elimination of investment in S Co as at acquisition date*

| | | |
|---|---|---|
| Dr Ordinary shares (S Co) | 6,500,000 | |
| Dr Opening retained earnings (S Co) | 1,170,000 | |
| Dr Goodwill | 5,421,000 | |
| Dr Inventories | 130,000 | |
| Dr Land | 390,000 | |
| Dr Buildings – net | 1,300,000 | |
| Cr Investment in S Co | | 13,260,000 |
| Cr Equipment – net | | 390,000 |
| Cr Contingent liabilities | | 130,000 |
| Cr Deferred tax liability (net) | | 260,000 |
| Cr Non-controlling interests | | 871,000 |

***Explanatory note:***

Goodwill recognized on consolidation purely relates to the parent's share. Non-controlling interests are measured as a proportion of identifiable net assets and goodwill attributable to non-controlling interests' share is not recognized.

*CJE2: Impairment of goodwill and amortization of excess of fair values of identifiable net assets recognized in the income statement (I/S) and statement of financial position (SFP)*

| | | |
|---|---|---|
| Dr Cost of sales (I/S) | 130,000 | |
| Dr Depreciation – buildings (I/S) | 26,000 | |
| Dr Impairment of goodwill (I/S) | 468,000 | |
| Dr Accumulated depreciation – equipment (SFP) | 78,000 | |
| Dr Contingent liabilities (SFP) | 130,000 | |
| Dr Tax expense (I/S) | 10,400 | |
| Cr Inventories (SFP) | | 130,000 |
| Cr Accumulated depreciation – buildings (SFP) | | 26,000 |
| Cr Goodwill (SFP) | | 468,000 |
| Cr Depreciation – equipment (I/S) | | 78,000 |
| Cr Contingent liability expense (I/S) | | 130,000 |
| Cr Deferred tax liability (net) (SFP) | | 10,400 |

*CJE3: Allocation of non-controlling's share of subsidiary's profits for the year*

| | | |
|---|---|---|
| Dr Income to non-controlling interests (I/S) | 76,960 | |
| Cr Non-controlling interests (SFP) | | 76,960 |

***Workings:***

| | |
|---|---|
| Net profit after tax of S Co for 20x2 | $728,000 |
| Increase in cost of sale of undervalued inventories | (130,000) |
| Increase in depreciation on undervalued building | (26,000) |
| Decrease in depreciation on overvalued equipment | 78,000 |
| Decrease in expense on earlier recognized contingency | 130,000 |
| Increase in tax expense as a result of the above adjustments | (10,400) |
| Adjusted net profit after tax of S Co | $769,600 |
| NCI's share of adjusted net profit after tax at 10% | $ 76,960 |

***Explanatory note:***

Since non-controlling interests' share of goodwill is not recognized, no adjustment is required for the impairment loss on goodwill.

Other consolidation adjusting entries in Illustration 4.7 apply and are not repeated here.

**Reconciliation of non-controlling interests with net assets of subsidiary**

First, reconstruct the non-controlling interests' balance from the consolidation adjustments.

| | | |
|---|---|---|
| Non-controlling interests at date of acquisition | $871,000 | (CJE1) |

| | | |
|---|---|---|
| Non-controlling interests' share of current profit after tax | 76,960 | (CJE3) |
| Less dividends declared to non-controlling interests . . | (45,500) | (CJE4) |
| Non-controlling interests as at 31 December 20x2 . . . | $902,460 | |



Next, determine the net assets or shareholders' equity of S Co. Non-controlling interests' balance in the statement of financial position should be equal to non-controlling interests' share of adjusted net assets or shareholders' equity of the subsidiary.

| | |
|---|---|
| Book value of shareholders' equity as at 31 December 20x2 (Note 1) . . . . . | $7,943,000 |
| Unamortized excess of fair value over book value at 31 December 20x2 (Note 2) . . . . | 1,081,600 |
| Adjusted shareholders' equity of S Co . . . . . | $9,024,600 |
| Non-controlling interests as at 31 December 20x2 . . . . . | $ 902,460 |

**Note 1:** Book value of shareholders' equity as at 31 December 20x2

| | | |
|---|---|---|
| Share capital . . . . . | | $6,500,000 |
| Retained earnings: | | |
| As at 1 January 20x2 . . . . . | $1,170,000 | |
| Net profit after tax . . . . . | 728,000 | |
| Dividends declared . . . . . | (455,000) | |
| As at 31 December 20x2 . . . . . | | 1,443,000 |
| Book value of shareholders' equity . . . . . | | $7,943,000 |

**Note 2:** Unamortized excess of fair value over book value at 31 December 20x2

| | | |
|---|---|---|
| Undervalued land . . . . . | $ 390,000 | |
| Undervalued buildings . . . . . | 1,274,000 | ($1,300,000 × 49/50) |
| Overvalued equipment . . . . . | (312,000) | ($390,000 × 4/5) |
| | $1,352,000 | |
| After-tax balance . . . . . | $1,081,600 | ($1,352,000 × 80%) |

*Undervalued inventories and contingent liabilities have a zero unamortized balance as at 31 December 20x2*

**2. Consolidation adjustments for 20x3**

*CJE1: Re-enactment of the elimination of investment in S Co as at acquisition date*

| | |
|---|---|
| Dr Ordinary shares . . . . . | 6,500,000 |
| Dr Opening retained earnings . . . . . | 1,170,000 |
| Dr Goodwill . . . . . | 5,421,000 |
| Dr Inventories . . . . . | 130,000 |

| | | |
|---|---|---|
| Dr Land | 390,000 | |
| Dr Buildings – net | 1,300,000 | |
| Cr Investment in S | | 13,260,000 |
| Cr Equipment – net | | 390,000 |
| Cr Contingent liabilities | | 130,000 |
| Cr Deferred tax liability (net) | | 260,000 |
| Cr Non-controlling interests | | 871,000 |



*CJE2: Goodwill impairment losses in 20x2*

| | | |
|---|---|---|
| Dr Opening retained earnings | 468,000 | |
| Cr Goodwill | | 468,000 |

***Explanatory note:***

Since goodwill is a parent asset, impairment losses are not shared with non-controlling interests. As goodwill impairment was debited to consolidated income statement in the previous year, we have to debit the opening retained earnings to reflect the prior-year consolidation entry.

Other consolidation adjusting entries in Illustration 4.7 apply and are not repeated here.

**Reconciliation of non-controlling interests with net assets of subsidiary**

*Non-controlling interests' balance reconstructed from the consolidation adjustments:*

| | | | |
|---|---|---|---|
| Non-controlling interests at date of acquisition | $ 871,000 | (CJE1) | |
| Non-controlling interests' share of post-acquisition retained earnings to beginning of the current year | 27,300 | (CJE3) | |
| Non-controlling interests' share of prior years' amortization of fair value adjustments | 4,160 | (CJE4) | |
| Non-controlling interests' share of current profit after tax | 12,960 | (CJE6) | |
| Non-controlling interests at 31 December 20x3 | $ 915,420 | | |
| Book value of shareholders' equity (Note 1) | | | $8,343,000 |
| Unamortized excess of fair value over book value at 31 December 20x3 (Note 2) | | | 811,200 |
| Adjusted shareholders' equity of S Co | | | $9,154,200 |
| Non-controlling interests' share | | | $915,420 |

**Note 1:** Book value of shareholders' equity as at 31 December 20x3

| | | |
|---|---|---|
| Share capital | | $6,500,000 |
| Retained earnings: | | |
| As at 1 January 20x3 | $1,443,000 | |
| Net profit after tax | 400,000 | |
| Dividends declared | 0 | |

| | |
|---|---|
| As at 31 December 20x3 | 1,843,000 |
| Book value of shareholders' equity | $8,343,000 |

**Note 2:** Unamortized excess of fair value over book value at 31 December 20x3

| | | |
|---|---|---|
| Undervalued buildings | $1,248,000 | ($1,300,000 × 48/50) |
| Overvalued equipment | (234,000) | ($390,000 × 3/5) |
| | $1,014,000 | |
| After-tax balance | $ 811,200 | ($1,014,000 × 80%) |

*(Undervalued inventories and land and contingent liabilities have a zero unamortized balance as at 31 December 20x3.)*



APPENDIX 4B

# Accounting for Other Components of Non-Controlling Interests

**ILLUSTRATION 4B.1**

On 1 Jan 20x1, P Co purchases all the 400,000 ordinary shares in S Co. The purchase consideration is $2,500,000 in cash. Net assets of S Co as at 1 Jan 20x1 amounted to $711,000. The fair value of the net identifiable assets as at 1 Jan 20x1 is $800,000. Tax effects are ignored for this illustration.

At the date of acquisition, S Co's 1,000 preference shares (with nominal value of $1 each) are classified as equity in accordance with IAS 32 *Financial Instruments: Presentation*. The preference shares do not carry any voting rights and will provide their holders a right to preferred dividends, which ranks higher than the payment of any dividend to ordinary shareholders. Upon liquidation, the preference shareholders are only entitled to receive the amount of $1 from the assets available for distribution in priority to the ordinary shareholders. They are not entitled to a proportionate share of the net assets of S Co on liquidation. The fair value of these preference shares at the date of acquisition is $2,000.

During the year, S Co has also issued share options to its employees. As at 1 Jan 20x1, two tranches of share options that has been issued and the carrying value in the share option reserve in equity amounted to $70,000. Tranche 1 that has a carrying value of $20,000 calculated in accordance with IFRS 2 *Share-Based Payment* has vested in the employees. Tranche 2 that has a carrying value of $50,000 has not vested in the employees as at the date of acquisition. The vesting period for Tranche 2 is two years and as of 1 Jan 20x1, the employees have completed one year out of the two-year vesting period. The market-based measure of the share option calculated in accordance with IFRS 2, assuming the acquisition date is the grant date, amounts to approximately $56,000. Both tranches of share

options do not expire on acquisition of S Co by P Co and P Co does not replace them on acquisition. There are also no changes to the terms to the share options.

During the year, S Co has also issued a convertible loan to third parties. The equity conversion option with carrying amount of $40,000 meets the criteria for equity classification and has been recognized as part of equity in the books of S Co. The fair value of the equity conversion option as at the date of acquisition of 1 Jan 20x1 amounted to $48,000.

**Analysis for components of non-controlling interests**

The preference shares, employee share options, and equity conversion option meet the definition of non-controlling interests in S Co as they represent equity in a subsidiary not attributable directly or indirectly to a P Co.



Analysis for preference shares

The preference shares do not carry equal rights and ranking of the ordinary shares issued by the Company. The preference shareholders do not carry voting rights. They are given priority in terms of a preferred dividend over the payment of any dividend to the ordinary shareholders and the holders are only entitled to only the nominal amount paid for the preference shares upon liquidation of S Co. Accordingly, they do not represent present ownerships interests that entitle their holders to a proportionate share of net assets upon liquidation and accordingly, the option to measure them at fair value or at their proportionate share in S Co's recognized amount of identifiable net assets is not available. IFRS 3 paragraph 19 requires them to be measured at fair value at the date of acquisition, which is $2,000.

Analysis for employee share option

*Tranche 1 – vested by the date of acquisition*

P Co does not replace the shares held by the employees of S Co on acquisition, and under the terms and conditions, the share options held by the employees do not expire on acquisition of S Co by P Co. The share options, while vested in the employees, do not represent present ownership interests that entitle their holders to a proportionate share of net assets upon liquidation, as the employees have not exercised their options to acquire the ordinary shares in the S Co as at the date of acquisition. Paragraph B62A of IFRS 3 requires these non-controlling interests that relate to the vested share options to be measured at their market-based measure in accordance with IFRS 2, which is $20,000.

*Tranche 2 – unvested by the date of acquisition*

Similarly, P Co does not replace the shares held by the employees of S Co on acquisition, and under the terms and conditions, the share options held by the employees do not expire on acquisition of S Co by P Co. The share options also do not represent present ownership interests that entitle their holders to a proportionate share of net assets upon liquidation as the share options have not vested, and accordingly, employees cannot exercise their options to acquire the ordinary shares in the S Co.

The difference between Tranches 1 and 2 lies in that the share options in Tranche 2 are not vested by the date of acquisition. As a result, the measurement basis differs. Specifically, Paragraphs B62A and B62B of IFRS 3 require these options to be measured at their market-based measure as if the acquisition date were the grant date, which in this case is $56,000. The amount allocated to the non-controlling interest relating to the share options in Tranche 2 is based on the ratio of the portion of the vesting period completed to the greater of the total vesting period and the original vesting period of the share-based payment transaction. Hence, in this case, the amount allocated to the non-controlling interest is $28,000 ($56,000 × 1 year/2years). The balance of $28,000 is allocated to post-combination service, that is, part of the $2,500,000 consideration transferred is deemed to be payment for this post-combination service. Effectively, the purchase consideration for the acquisition is $2,472,000 ($2,500,000 − $28,000).

Analysis for equity conversion option

The equity conversion option that are held by the external parties, that is, owners of the convertible loan do not represent present ownership interests, which entitle their holder to a proportionate share of net assets upon liquidation. The options have yet to be exercised by the holders as at the date of acquisition in exchange for ordinary shares. Accordingly, IFRS 3 paragraph 19 requires them to be measured at fair value of $48,000 at the date of acquisition.

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**Calculation of fair value uplift on acquisition (excess of fair value over book value of identifiable net assets)**

$$\begin{aligned}\text{Fair value uplift on acquisition} &= \$800{,}000 - \$711{,}000\\ &= \$89{,}000\end{aligned}$$

**Calculation of goodwill**

$$\begin{aligned}\text{Consideration transferred} &= \$2{,}500{,}000 - \$28{,}000 \text{ (being payment for post-combination employee service)}\\ &= \$2{,}472{,}000\end{aligned}$$

$$\begin{aligned}\text{Goodwill} &= \text{Fair value of consideration transferred} + \text{Amount of non-controlling interests at acquisition date} - \text{Recognized net identifiable assets of acquiree measured in accordance with IFRS 3}\\ &= \$2{,}472{,}000 \quad \left[\begin{array}{l}+\ \$2{,}000 + \$20{,}000 + \$28{,}000\\ +\ \$48{,}000\end{array}\right] - \$800{,}000\\ &= \$1{,}770{,}000\end{aligned}$$

**Preparation of the consolidation journal entries:**

| | | |
|---|---|---|
| Dr Share capital and other reserves | 711,000 | |
| Dr Net identifiable assets (Excess of fair value over NBV) | 89,000 | |
| Dr Goodwill | 1,770,000 | |
| Dr Staff costs – P/L | 28,000 | |
| Cr Investment in subsidiary | | 2,500,000 |
| Cr NCI - Preference shares | | 2,000 |
| Cr NCI - Share option reserve | | 48,000 |
| Cr NCI - Equity component of convertible loan | | 48,000 |

*(Being elimination of share capital and investment account.)*

***Explanatory note:***

Included in the debit entry for share capital and other reserves are balances recorded in the books of the subsidiary for preference shares, share option reserve, and equity component of convertible loan. These amounts recorded in the subsidiary's books are eliminated so that separate credit entries can be made to record the fair values of these items in the consolidated financial statements.

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# CONCEPT QUESTIONS

**State True or False and the reasons for your response.**

**CQ4.1** The book value of shares issued in exchange for control rights should be used as the basis of measuring the initial cost.

**CQ4.2** Goodwill is the difference between consideration transferred and the share of fair value of the net assets of a subsidiary.

**CQ4.3** Goodwill impairment losses in previous years will have to be credited to opening retained earnings in the consolidation workpapers.

**CQ4.4** A group may comprise a parent, its subsidiaries, and the subsidiaries of its subsidiaries.

**CQ4.5** The consolidation adjustment to eliminate the parent's investment and the subsidiary's equity at the date of acquisition is passed once in the first year of acquisition and not subsequently.

**CQ4.6** The adjustment in (5) above, that is, the elimination entry as at the date of acquisition is an "asset substitution" entry. It transforms the investment asset into goodwill and the fair value of identifiable assets and liabilities of the subsidiary.

**CQ4.7** Non-controlling interests' share of a subsidiary's retained earnings from the date of acquisition to the beginning of the current period will result in a debit entry to non-controlling interests and a credit entry to retained earnings.

**CQ4.8** Non-controlling interests of profitable subsidiaries result in a credit item in the income statement and a debit balance in the statement of financial position.

**CQ4.9** Dividends declared by a group will include dividends declared by the parent and dividends declared by a subsidiary to its non-controlling interests.

**CQ4.10** It is possible for a group to show a consolidated net profit after tax that is smaller than the consolidated profit retained by the group.

**CQ4.11** Equity components, which are not present ownership interests and do not entitle their holders to a pro-rata share of net assets during liquidation, do not qualify as non-controlling interests under IFRS 3.

**CQ4.12** Equity components, which are not present ownership interests and do not entitle their holders to a pro-rata share of net assets, are measured at nil balance at the date of acquisition.



# RESEARCH QUESTIONS

**RQ4.1 Accounting choices on accounting for non-controlling interests**

IFRS 3 *Business Combinations* allows a choice on the basis of measurement for non-controlling interests. For each acquisition, the acquirer may choose to measure non-controlling interests on acquisition date at either fair value or as a proportion of identifiable net assets. The basis to measure non-controlling interests as a proportion of identifiable net assets retains the accounting treatment of the previous standard IAS 22 *Business Combinations*. This accounting choice puts IFRS 3 at variance with US GAAP.

***Required:***

1. Choose a sample of any five companies that applies IFRS 3 *Business Combinations* and note the basis of measurement for non-controlling interests for each of the five companies.
2. Explain the directional impact of each basis on the following measures:
   (a) Total assets
   (b) Total equity
   (c) Net profit after tax
   (d) Return on equity
   (e) Debt/Equity ratio
   (f) Sales/Assets ratio
3. Choose one of the companies that you had chosen in Question 1. Assuming that you are a shareholder or a potential investor of that company, which basis of measurement would provide you with the information that you require? Explain.
4. Choose one of the companies that you had chosen in Question 1. Assuming that you are a consultant to that company and ignoring their existing choice, advise that company on the choice of accounting basis.

### RQ4.2 Debit balance of non-controlling interests

IFRS 10 *Consolidated Financial Statements* requires the allocation of profit to non-controlling interests even if the allocation would result in a debit balance of non-controlling interests. This requirement is different from IAS 27 (2003) *Consolidated and Separate Financial Statements*, which requires the majority interests to absorb the losses that exceed the then "minority interests." IAS 27 (2003) does permit an exception: if the "minority interests" have a binding obligation and is able to make an additional investment to make good the losses, the majority interests do not need to absorb the losses covered by the obligation.

***Required:***

1. Read the Basis of Conclusions in IFRS 10 and analyze the explanations given by the Board on their final decision. (Most University Libraries would have a database that would permit you to access the Basis of Conclusions of the IFRS Standards.)
2. When losses exceed the balance in non-controlling interests, which of the following treatment would you support. Write a 500-word short essay to explain your decision.
   (a) IFRS 10;
   (b) IAS 27 (2003); or
   (c) Other treatment (specify).
3. *Role play and class debate*: Your professor will organize you into two groups. One group represents the board members in the Board that supports the position in IFRS 10, and the other group supports the position in IAS 27 (2003). Each group is given ten minutes to make its case. The professor and other class members will be the judge to decide which group has the most convincing arguments.

# PROBLEMS

### P4.1 Goodwill fair value adjustments

P Co issued 2,000,000 of its own shares (fair value of $10 per share) and paid $6,000,000 in cash to the existing owners of S Co to acquire 90% of the shares of S Co on 1 July 20x1. Fair value of non-controlling interests was $2,800,000 at the date of acquisition. The book and fair values of S Co's assets and liabilities as at 1 July 20x1 are shown below:

| | Book value | Fair value |
|---|---|---|
| Plant and equipment | $ 2,000,000 | $ 1,800,000 |
| Investment property | 10,000,000 | 15,000,000 |
| In-process R&D | 0 | 6,000,000 |
| Inventory | 500,000 | 750,000 |
| Accounts receivable | 200,000 | 200,000 |
| Cash | 10,000 | 10,000 |
| | $12,710,000 | $23,760,000 |
| Accounts payable | $ 1,510,000 | $ 1,510,000 |
| Contingent liabilities | | 90,000 |
| Share capital | 10,000,000 | |
| Retained earnings | 1,200,000 | |
| | $12,710,000 | |

***Additional information:***

(a) The remaining useful life of plant and equipment as at 1 July 20x1 was ten years.

(b) As at 30 June 20x2, the fair value of in-process R&D was reliably assessed at $5,500,000.

(c) 90% of the inventory was sold by 30 June 20x2 and the balance 10% was deemed as impaired on 30 June 20x3.

(d) The fair value model is to be adopted for investment property in the consolidated financial statements, that is, changes in fair value after initial recognition are taken to the income statement. As at 30 June 20x2, the fair value of the investment property was $16,000,000. S Co incorrectly applied the cost model (without depreciation) to measure the investment property in its separate financial statements.

(e) As at 30 June 20x2, no recognition was made for contingent liabilities in the separate financial statements of S Co. However, a disclosure was made in the footnotes of S Co in relation to the contingent liabilities. As at 30 June 20x2, the reported amount of the contingent liabilities was deemed reliable and met the recognition criteria in IFRS 3.

(f) Approximately 10% of entity goodwill was deemed to be impaired as of 30 June 20x2.

(g) Tax rate is 20%. Recognize tax effects on fair value adjustments.

(h) S Co earned annual profit after tax of $2,000,000 for the year ended 30 June 20x2 and 30 June 20x3. There were no dividends or other changes in equity during the two years. There was no change in the fair value of investment property as at 30 June 20x3.

***Required:***

1. Show the consolidation adjustments for P Co and its subsidiary S Co for the years ended 30 June 20x2 and 30 June 20x3.
2. Perform an analytical check on the balance of non-controlling interests as at 30 June 20x2 and 30 June 20x3.

### P4.2 Consolidation, goodwill, and fair value adjustments

Prince Ltd bought 80% of Silver Ltd on 1 January 20x1 for $230,000 when Silver's statement of financial position was as follows:

**Silver Ltd**
**Statement of Financial Position**
**As at 1 January 20x1**

| | Book value | Fair value |
|---|---|---|
| Share capital | $190,000 | |
| Retained earnings | 5,000 | |
| Shareholders' equity | $195,000 | $215,000 |
| Fixed assets, cost | $100,000 | |
| Less accumulated depreciation | 30,000 | |
| Net book value | $ 70,000 | $ 90,000 |
| Other assets, net of liabilities | 125,000 | 125,000 |
| Net assets | $195,000 | $215,000 |

Fixed assets had a remaining useful life of five years as at 1 January 20x1. Goodwill impairment losses on the original goodwill of Silver attributable to parent and non-controlling interests are as follows:

(a) 20% of the original goodwill was deemed impaired and written off in 20x2.
(b) 10% of the original goodwill was written off in 20x3.

Fair value of non-controlling interests as at 1 January 20x1 was $55,000. The financial statements for the year ended 31 December 20x3 are as follows:



**Income Statement and Partial Statement of Changes in Equity**
**For Year Ended 31 December 20x3**

| | Prince Ltd | Silver Ltd |
|---|---|---|
| Sales | $5,000,000 | $1,900,000 |
| Cost of sales | (4,250,000) | (1,520,000) |
| Gross profit | $ 750,000 | $ 380,000 |
| Other expenses | (185,000) | (155,000) |
| Operating profit | $ 565,000 | $ 225,000 |
| Dividend income from Silver Ltd | 28,000 | 0 |
| Profit before tax | $ 593,000 | $ 225,000 |
| Tax expense | (113,000) | (45,000) |
| Profit after tax | $ 480,000 | $ 180,000 |

| | | |
|---|---|---|
| Dividends declared | (100,000) | (35,000) |
| Profit retained | $ 380,000 | $ 145,000 |
| Retained earnings, 1 January | 1,620,000 | 155,000 |
| Retained earnings, 31 December | $2,000,000 | $ 300,000 |

**Statement of Financial Position**
**As at 31 December 20x3**

| | Prince Ltd | Silver Ltd |
|---|---|---|
| Fixed assets, net book value | $2,200,000 | $326,000 |
| Investment in Silver Ltd, at cost | 230,000 | |
| Other investments | | 120,000 |
| Inventories | 797,000 | 106,000 |
| Receivables | 453,000 | 50,000 |
| Due from Silver Ltd | 60,000 | |
| Cash | 185,000 | 20,000 |
| | $3,925,000 | $622,000 |
| Share capital | $1,150,000 | $190,000 |
| Retained earnings | 2,000,000 | 300,000 |
| Payables | 775,000 | 72,000 |
| Due to Prince Ltd | | 60,000 |
| | $3,925,000 | $622,000 |

***Required:***

1. Prepare all necessary consolidation adjustments and elimination entries for the year ended 31 December 20x3. Recognize tax effects at 20%.
2. Prepare the consolidation worksheet for the year ended 31 December 20x3.
3. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x3.



### P4.3 Consolidation and analytical check on non-controlling interests

The statement of financial position of Subsidiary Co as at 31 December 20x0, the date it was acquired by Parent Co, and income statements for Parent Co and Subsidiary Co for the year ended 31 December 20x1 are shown below:

**Subsidiary Co**
**Statement of Financial Position**
**As at 31 December 20x0 (date of acquisition)**

| | Book value | Fair value |
|---|---|---|
| Fixed assets, net | $600,000 | $750,000 |
| Current assets: | | |
| Inventory | 40,000 | 50,000 |

| | | |
|---|---|---|
| Accounts receivable | 100,000 | 70,000 |
| Cash | 20,000 | 20,000 |
| | $160,000 | $140,000 |
| Less current liabilities: | | |
| Accounts payable | $ 80,000 | $ 80,000 |
| Short-term loans | 110,000 | 110,000 |
| | $190,000 | $190,000 |
| Net current assets/liabilities | (30,000) | (50,000) |
| | $570,000 | $700,000 |
| Shareholders' equity: | | |
| Share capital | $320,000 | |
| Retained earnings | 250,000 | |
| | $570,000 | $700,000 |

**Income Statement and Partial Statement of Changes in Equity**
**For Year Ended 31 December 20x1**

| | Parent Co | Subsidiary Co |
|---|---|---|
| Net profit before tax (including dividend income) | $ 550,000 | $100,000 |
| Less tax expense | 110,000 | 20,000 |
| Net profit after tax | $ 440,000 | $ 80,000 |
| Less dividends declared | 100,000 | 24,000 |
| Profit retained for the year | $ 340,000 | $ 56,000 |
| Retained earnings, 1 January | 1,000,000 | 250,000 |
| Retained earnings, 31 December | $1,340,000 | $306,000 |



***Additional information:***

(a) Parent paid $1,000,000 for an 80% stake in Subsidiary on 31 December 20x0. Fair value of non-controlling interests was $250,000 at acquisition date that was proportional to the consideration transferred by Parent Co.

(b) On 31 December 20x1, total goodwill was assessed to be impaired to the extent of $50,000.

(c) There were no intragroup transactions other than Parent Co's investment in Subsidiary Co and dividends declared by S Co.

(d) Information relating to "undervalued" and "overvalued" identifiable assets is as follows:

  (i) Remaining useful life of undervalued fixed asset as at 31 December 20x0 was ten years.

  (ii) Undervalued inventory was sold in 20x1.

  (iii) Overvalued accounts receivable was written down for a potential bad debt. The debt was confirmed bad in 20x1 and written off in Subsidiary's books in 20x1.

(e) Assume a tax rate of 20%. Recognize tax effects on fair value adjustments.

***Required:***

1. What is the goodwill arising from the application of the requirements of IFRS 3 *Business Combinations*?
2. Prepare the consolidation adjustments for the year ended 31 December 20x1.
3. Perform an analytical check on the non-controlling interests' balance as at 31 December 20x1.
4. Show the consolidation adjustments that need to be passed in 20x2 to re-enact the consolidation adjustments of 20x1.
5. Prepare the consolidated income statement for the year ended 31 December 20x1.

### P4.4 Consolidation and analytical check on non-controlling interests

On 1 January 20x4, P Co acquired 90% of S Co. Details of S Co as at the date of acquisition are as follows:

| | |
|---|---|
| Share capital | $1,000,000 |
| Retained earnings as at 1 January 20x4 | 150,000 |
| | $1,150,000 |

Fair value was different from book value of the following assets of S Co:

| | Fair value | Book value |
|---|---|---|
| Fixed assets | $900,000 | $800,000 |

Remaining useful life for the fixed assets as at acquisition date was five years and residual value of the fixed assets was zero. Fair value of non-controlling interests as at acquisition date was $345,000. The financial statements of P Co and S Co for the year ended 31 December 20x5 are shown below. Assume a tax rate of 20% throughout.



**Income Statement and Partial Statement of Changes in Equity**
**For Year Ended 31 December 20x5**

| | P Co | S Co |
|---|---|---|
| Sales | $10,000,000 | $3,500,000 |
| Cost of sales | (7,820,000) | (1,200,000) |
| Gross profit | $ 2,180,000 | $2,300,000 |
| Operating expenses | (800,000) | (560,000) |
| Operating profit | $ 1,380,000 | $1,740,000 |
| Dividend income from S | 54,000 | |
| Interest income – S and bank | 100,000 | |
| Interest expense to P | | (90,000) |
| Profit before tax | $ 1,534,000 | $1,650,000 |
| Tax at 20% | (306,800) | (330,000) |
| Profit after tax | $ 1,227,200 | $1,320,000 |
| Dividends declared | (120,000) | (60,000) |
| Profit retained | $ 1,107,200 | $1,260,000 |

| | | |
|---|---|---|
| Retained earnings, 1 January | 1,300,000 | 400,000 |
| Retained earnings, 31 December | $ 2,407,200 | $1,660,000 |

**Statement of Financial Position**
**As at 31 December 20x5**

| | P Co | S Co |
|---|---|---|
| Investment in S Co | $ 3,200,000 | |
| Fixed assets | 4,500,000 | $2,000,000 |
| Loan receivable from S Co | 500,000 | |
| Inventory | 1,250,000 | 670,000 |
| Accounts receivable | 1,300,000 | 500,000 |
| Other receivables | 200,000 | 120,000 |
| Cash | 320,000 | |
| | $11,270,000 | $3,290,000 |
| Share capital | $ 3,000,000 | $1,000,000 |
| Retained earnings | 2,407,200 | 1,660,000 |
| Loan payable to P | | 500,000 |
| Accounts payable | 5,742,800 | 80,000 |
| Other payables | 120,000 | 50,000 |
| | $11,270,000 | $3,290,000 |

***Required:***

1. Prepare the consolidation adjustments for the year ended 31 December 20x5.
2. Prepare the consolidation worksheets for the year ended 31 December 20x5.
3. Reconcile the non-controlling interests' balance as at 31 December 20x5.



### P4.5 Acquisition method and non-controlling interests at fair value

P Co acquired control of Jasper Co through acquisition of 90% in the voting rights of Jasper Co on 1 July 20x2. A cash transfer of $2,000,000 was made to the former owners of Jasper Co. P Co elects to measure non-controlling interests at fair value on acquisition date. The fair value of non-controlling interests on 1 July 20x2 was $200,000. The shareholders' equity of Jasper Co at acquisition date is as follows:

| | |
|---|---|
| Share capital | $ 500,000 |
| Retained earnings | 450,000 |
| Revaluation reserves | 100,000 |
| | $1,050,000 |

The fair and book values of identifiable net assets of Jasper Co at acquisition date is shown below:

| | Book value | Fair value | Remarks |
|---|---|---|---|
| Inventory | $ 200,000 | $ 250,000 | 60% were sold during 20x3 |
| Fixed assets | 450,000 | 500,000 | Remaining useful life at acquisition was 5 years. |
| Other net assets | 400,000 | 400,000 | |
| | $1,050,000 | $1,150,000 | |

The financial statements of P Co and Jasper Co are shown below:

**Statement of Comprehensive Income**
**For Year Ended 31 December 20x3**

| | P Co | Jasper Co |
|---|---|---|
| Profit before tax | $4,000,000 | $1,000,000 |
| Tax | (800,000) | (200,000) |
| Profit after tax | $3,200,000 | $ 800,000 |
| Change in revaluation reserves, after-tax | | 100,000 |
| Comprehensive income | $3,200,000 | $ 900,000 |

**Statement of Changes in Equity**
**For the Year Ended 31 December 20x3 (partial)**

| | P Co | Jasper Co | Jasper Co | Jasper Co |
|---|---|---|---|---|
| | Retained earnings | Retained earnings | Revaluation reserves | Total |
| Balance, 1 January 20x3 | $ 840,000 | $ 400,000 | $400,000 | $ 800,000 |
| Comprehensive income | 3,200,000 | 800,000 | 100,000 | 900,000 |
| Dividends declared | (180,000) | (100,000) | | (100,000) |
| Balance, 31 December 20x3 | $3,860,000 | $1,100,000 | $500,000 | $1,600,000 |



**Statement of Financial Position**
**As at 31 December 20x3**

| | P Co | Jasper Co |
|---|---|---|
| Fixed assets, net book value | $2,400,000 | $2,200,000 |
| Investment in Jasper Co | 2,000,000 | |
| Inventory | 720,000 | 500,000 |
| Intercompany receivable | | 300,000 |
| Accounts receivable | 800,000 | 550,000 |
| Dividend receivable from Jasper | 72,000 | |
| Cash | 60,000 | 20,000 |
| | $6,052,000 | $3,570,000 |
| Payables | $ 712,000 | $1,390,000 |

| | | |
|---|---|---|
| Intercompany payable | 300,000 | |
| Dividend payable | 180,000 | 80,000 |
| Share capital | 1,000,000 | 500,000 |
| Retained earnings | 3,860,000 | 1,100,000 |
| Revaluation reserves | | 500,000 |
| | $6,052,000 | $3,570,000 |

***Required:***

1. Prepare the consolidation adjusting entries for the year ended 31 December 20x3. Tax rate was 20%. Recognize tax effects on fair value differentials.
2. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x3.

### P4.6 Acquisition method and non-controlling interests as a proportion of identifiable net assets

Assume the same facts as in P4.5, except that P Co elects to measure non-controlling interests as a proportion of identifiable net assets.

***Required:***

1. Prepare the consolidation adjusting entries for the year ended 31 December 20x3. Tax rate was 20%. Recognize tax effects on fair value differentials.
2. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x3.

### P4.7 Acquisition method and non-controlling interests at fair value

P Co acquired a controlling interest in Moonstone as follows:

| | |
|---|---|
| Interest acquired | 90% |
| Date of acquisition | 1 January 20x1 |



At acquisition date, the shareholder's equity of Moonstone Co were as follows:

| | |
|---|---|
| Share capital | $1,200,000 |
| Retained earnings | 550,000 |
| Revaluation reserves | 120,000 |
| | $1,870,000 |

The fair and book values of identifiable net assets of Moonstone at date of acquisition were as follows:

| | Book value | Fair value | Remarks |
|---|---|---|---|
| Inventory | $ 120,000 | $ 180,000 | Sold by Moonstone Co during 20x1 |
| Intangible assets | – | 250,000 | Remaining useful life was five years |
| Other net assets | $1,750,000 | 1,750,000 | |
| | $1,870,000 | $2,180,000 | |

P Co chose to measure non-controlling interests at fair value on acquisition date. The fair value of non-controlling interests in Moonstone Co as at date of acquisition was $250,000. The financial statements of P Co and Moonstone Co are shown below:

**Statement of Comprehensive Income**
**For the year ending 31 December 20x3**

| | P Co | Moonstone Co |
|---|---|---|
| Profit before tax | $2,000,000 | $500,000 |
| Tax | (391,000) | (100,000) |
| Profit after tax | $1,609,000 | $400,000 |
| Other comprehensive income: | | |
| Change in revaluation reserve, after-tax | | 90,000 |
| Comprehensive income | $1,609,000 | $490,000 |

**Statement of Changes in Equity**
**For the year ended 31 December 20x3 (partial)**

| | P Co | Moonstone | Moonstone | Moonstone |
|---|---|---|---|---|
| | Retained earnings | Retained earnings | Revaluation reserves | Total |
| Balance, 1 January 20x3 | $ 760,000 | $520,000 | $200,000 | $ 720,000 |
| Comprehensive income | 1,609,000 | 400,000 | 90,000 | 490,000 |
| Dividends declared | (100,000) | (50,000) | | (50,000) |
| Balance, 31 December 20x3 | $2,269,000 | $870,000 | $290,000 | $1,160,000 |



**Statement of Financial Position**
**As at 31 December 20x3**

| | P Co | Moonstone Co |
|---|---|---|
| Fixed assets, net book value | $2,500,000 | $1,800,000 |
| Investment in Moonstone Co | 2,600,000 | |
| Inventory | 900,000 | 500,000 |
| Intercompany receivable | | 120,000 |
| Accounts receivable | 1,000,000 | 500,000 |
| Dividend receivable from Moonstone | 45,000 | |
| Cash | 80,000 | 150,000 |
| | $7,125,000 | $3,070,000 |
| Payables | $2,706,000 | $ 660,000 |
| Intercompany payable | 120,000 | |
| Dividend payable | 30,000 | 50,000 |

| | | |
|---|---|---|
| Share capital | 2,000,000 | 1,200,000 |
| Retained earnings | 2,269,000 | 870,000 |
| Revaluation reserves | | 290,000 |
| | $7,125,000 | $3,070,000 |

***Required:***

1. Prepare the consolidation adjusting entries for the year ending 31 December 20x3. Tax rate was 20%. Recognize tax effects on fair value differentials.
2. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x3.

### P4.8 Acquisition method and non-controlling interests as a proportion of identifiable net assets

Assume the same facts as in P4.7, except that P Co elects to measure non-controlling interests as a proportion of identifiable net assets.

***Required:***

1. Prepare the consolidation adjusting entries for the year ended 31 December 20x3. Tax rate was 20%. Recognize tax effects on fair value differentials.
2. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x3.

### P4.9 Potential voting rights and non-controlling interests at fair value

P Co acquired control of Sapphire Co on 1 July 20x8 by acquiring 90% of the ordinary shares and stock options of Sapphire Co through paying cash of $200,000,000 to the former owners of Sapphire Co. P Co measures non-controlling interests at fair value. Tax on fair value of identifiable net assets should be recognized at 20%. The information on identifiable net assets of Sapphire Co is shown below:



| | Fair value | Book value |
|---|---|---|
| Intangible assets | $ 20,000,000 | $ 0 |
| Other net assets | 120,000,000 | 120,000,000 |
| Total net assets | $140,000,000 | $120,000,000 |

| | |
|---|---|
| Fair value per ordinary share | $2.00 |
| Fair value per stock option | $0.50 |
| Non-controlling interests | 10,000,000 shares; 1,000,000 option units |
| P Co's interests | 90,000,000 shares; 9,000,000 option units |
| Fair value of consideration transferred | $200,000,000 |

***Required:***

1. Determine the amount of non-controlling interests that has to be recognized by P Co as at acquisition date.
2. Determine the amount of goodwill that has to be recognized by P Co as at acquisition date.

### P4.10 Potential voting rights and non-controlling interests as a proportion of identifiable net assets

Assume the same facts as in P4.9, except that P Co measures non-controlling interests as a proportion of identifiable net assets. Assume that the ordinary shares and stock options share proportionately in the fair value of identifiable net assets.

***Required:***

1. Determine the amount of non-controlling interests that has to be recognized by P Co as at acquisition date.
2. Determine the amount of goodwill that has to be recognized by P Co as at acquisition date.

### P4.11 Consolidation and analytical check on non-controlling interests.

P Co acquired an ownership interest of 90% in X Co and obtained control on 1 January 20x3.

| Shareholders' equity of X Co at date of acquisition was as follows: | |
|---|---|
| Share capital | $600,000 |
| Retained earnings | 600,000 |
| | $1,200,000 |

On acquisition date, the fair value of fixed assets of X Co was $1,400,000 while the book value of fixed assets was $1,000,000. Fixed assets had a useful life of ten years as at the date of acquisition. The fixed assets were disposed at a price of $300,000 to third parties on 1 January 20x6.

Fair value of non-controlling interests of X Co as at acquisition date was $180,000. The financial statements for the year ended 31 December 20x6 are shown below.



**Income Statement and partial Statement of Changes in Equity for year ended 31 December 20x6**

| | P Co | X Co |
|---|---|---|
| Profit before tax | $4,200,000 | $2,400,000 |
| Tax | (840,000) | (480,000) |
| Profit after tax | $3,360,000 | $1,920,000 |
| Dividends declared | (200,000) | (100,000) |
| Profit retained | $3,160,000 | $1,820,000 |
| Retained earnings, 1 Jan 20x6 | 1,500,000 | 800,000 |
| Retained earnings, 31 Dec 20x6 | $4,660,000 | $2,620,000 |

**Statement of Financial Position as at 31 December 20x6**

| | P Co | X Co |
|---|---|---|
| Fixed assets, net book value | $5,200,000 | $2,600,000 |
| Investment in X, at cost | 1,800,000 | |
| Inventory | 300,000 | 340,000 |
| Accounts receivable | 500,000 | 250,000 |

| | | |
|---|---|---|
| Cash | 100,000 | 70,000 |
| *Total assets* | $7,900,000 | $3,260,000 |
| Payables | $2,240,000 | $40,000 |
| Share capital | 1,000,000 | 600,000 |
| Retained earnings | 4,660,000 | 2,620,000 |
| *Equity and Liabilities* | $7,900,000 | $3,260,000 |

Assume a tax rate of 20%. Recognize tax on fair value adjustments.

***Required:***

1. Prepare consolidation entries for the year ended 31 December 20x6.
2. Perform an analytical check on the balance in non-controlling interests as at 31 December 20x6, showing the workings clearly.

### P4.12 Consolidation and analytical check on non-controlling interests

P Co obtained control of Y Co on 1 January 20x4 by acquiring a 90% interest in Y Co. Share capital of Y Co was $900,000 and its retained earnings at date of acquisition was $300,000. Fair value of non-controlling interests of Y Co on 1 January 20x4 was $120,000.

Due diligence carried out before acquisition revealed that Y Co was a defendant in a litigation case in which a judgement was made in favor of the plaintiff. As Y Co is countersuing the plaintiff, no provision had yet been recognized as at 1 January 20x4 as the loss was not deemed probable. Based on expected losses, a provision of $200,000 should be recognized by P Co in accordance with IFRS 3. Y Co recognized a litigation expense of $180,000 in July 20x5 on the final award of damages to the plaintiff by the courts.



The financial statements of P Co and Y Co for the current year ended 31 December 20x6 are shown below.

**Income Statement and partial Statement of Changes in Equity for year ended 31 December 20x6**

| | P Co | Y Co |
|---|---|---|
| Profit before tax | $5,000,000 | $2,000,000 |
| Tax | (1,000,000) | (400,000) |
| Profit after tax | $4,000,000 | $1,600,000 |
| Dividends declared | (300,000) | (120,000) |
| Profit retained | $3,700,000 | $1,480,000 |
| Retained earnings, 1 Jan 20x6 | 1,200,000 | 800,000 |
| Retained earnings, 31 Dec 20x6 | $4,900,000 | $2,280,000 |

**Statement of Financial Position as at 31 December 20x6**

| | P Co | Y Co |
|---|---|---|
| Fixed assets, net book value | $4,160,000 | $2,500,000 |
| Investment in Y co, at cost | 1,800,000 | |

| | | |
|---|---|---|
| Amount due from P Co | | 90,000 |
| Inventory | 800,000 | 600,000 |
| Accounts receivable | 520,000 | 200,000 |
| Cash | 100,000 | 50,000 |
| *Total assets* | $7,380,000 | $3,440,000 |
| Accounts payable | $390,000 | $ 260,000 |
| Amount due to Y Co | 90,000 | |
| Share capital | 2,000,000 | 900,000 |
| Retained earnings | 4,900,000 | 2,280,000 |
| *Equity and Liabilities* | $7,380,000 | $3,440,000 |

***Required:***

1. Prepare consolidation entries for the year ended 31 December 20x6.
2. Perform an analytical check on the balance in non-controlling interests as at 31 December 20x6, showing the workings clearly.

**P 4.13 Business combinations, consolidation and analytical check on non-controlling interests**

On 2 January 20x4, P Co purchased from X Co, a controlling interest of 90% interest in Sapphire Co, a company that has an on-going research and development (R&D) project. On 2 January 20x4, P Co entered into the following transactions relating to the acquisition and financing arrangements.



| | | |
|---|---|---|
| (1) | Number of P Co's shares issued to X Co | 1,300,000 |
| | Fair value per share of P Co | $1.40 |
| (2) | Direct costs of issuing shares to X Co | $17,800 |
| (3) | Transfer of land to X Co | |
| | Fair value of land | $1,500,000 |
| | Carrying amount in P Co's books | $1,350,000 |
| (4) | Cash transferred to X Co | $ 800,000 |
| (5) | Number of shares issued to existing owners of P Co to fund the acquisition | 1,200,000 |
| (6) | Conditional payment made to X Co if R&D project is successful at the end of 3 years from acquisition date | $1,800,000 |
| | Probability of success | 70% |
| | Payment is due at the end of 3 years | 31 Dec 20x6 |
| | Implicit interest rate charged by X Co | 5% p.a. |

At the date of acquisition, the shareholders' equity of Sapphire Co was as follows:

| | |
|---|---|
| Share capital | $1,000,000 |
| Retained earnings | 158,000 |

| | |
|---|---|
| Other comprehensive income | 24,000 |
| | $1,182,000 |

The fair value of non-controlling interests on acquisition date was $412,000.

As at acquisition date, Sapphire Co had a research and development project that had the following expected outcomes:

| Future event | Present value of cash inflows | Probability |
|---|---|---|
| Successful outcome | $2,000,000 | 0.70 |
| Not successful outcome | $0 | 0.30 |

Prior to acquisition, Sapphire had accounted for the project expenditures as follows:

| | |
|---|---|
| Expensed research costs | $650,000 |
| Expensed development costs | $200,000 |
| Capitalized development costs as at 31 Dec 20x3 | $350,000 |

The development expenditures incurred to acquisition date of $350,000 met the conditions for capitalization in International Accounting Standard (IAS) 38 *Intangible Assets* and was recognized in Sapphire's balance sheet at acquisition date.



Sapphire Co completed the research and development project on 31 December 20x5 successfully. The estimated economic life of the intangible asset was 10 years. The group and Sapphire Co use the cost model to measure intangible assets.

The financial statements of Sapphire Co for the current year ended 31 December 20x6 are as follows:

**Income Statement and Statement of Changes in Equity (partial)**
**for the year ended 31 December 20x6**

| | |
|---|---|
| Profit before tax | $650,000 |
| Tax | (130,000) |
| Profit after tax | $520,000 |
| Dividends declared | (67,000) |
| Profit retained | $453,000 |
| Retained earnings, 1 Jan 20x6 | 239,000 |
| Retained earnings, 31 Dec 20x6 | $692,000 |

**Statement of Financial Position as at 31 Dec 20x6**

| | |
|---|---|
| Non-current assets | $1,560,000 |
| Amount due from P Co | 79,000 |
| Inventory | 237,000 |
| Accounts receivable | 270,000 |

| | |
|---|---|
| Cash | 32,000 |
| | $2,178,000 |
| Liabilities | $396,000 |
| Share capital | 1,000,000 |
| Retained earnings | 692,000 |
| Other comprehensive income balance | 90,000 |
| | $2,178,000 |

***Additional information:***

(a) Recognize tax effects on fair value adjustments and other adjustments at the tax rate of 20%.

(b) The balance of other comprehensive income as at 1 January 20x6 was $78,000. Other comprehensive income (OCI) arises from items of income that bypass net income in accordance with IAS 1 Presentation of Financial Statements. OCI is built up in separate accounts in equity. (Hint: Where OCI balance at beginning of the year is provided, separate allocation of current and change in post-acquisition OCI in the same way as for profit and post-acquisition retained earnings).

***Required:***

1. Prepare the journal entries to record the investment in Sapphire, transaction costs, financing transactions and other entries in P Co's books on 2 January 20x4. Recognize unamortized discount separately for future settlements.
2. In the light of the successful completion of the research and development project by Sapphire, record the following transactions in P Co's books for the year ended 31 December 20x5 and 20x6: page 234
   i. Interest expense on contingent consideration payable; and
   ii. Adjusting entries and settlement of contingent consideration payable.
3. Prepare consolidation entries for the year ended 31 December 20x6, with narratives (brief headers) and workings in accordance with IFRS 3 and IFRS 10.
4. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x6, showing the workings clearly.

### P4.14 Business combinations, consolidation and analytical check on non-controlling interests

On 2 January 20x4, P Co acquired controlling interest in 90% of Topaz Co through the following transfers to Y Co, the former owners of Topaz. P Co expects significant synergies from Topaz Co's possible licensing rights expected in 20x5. Financing and other transactions are also shown below:

| | |
|---|---|
| (1) Number of P Co shares issued to Y Co | 600,000 |
| Fair value per share of P Co | $1.80 |
| (2) Direct cost of issuing P Co shares | $20,000 |
| (3) Due diligence costs paid to consultants | $25,000 |
| (4) Cash transferred to Y Co | $800,000 |
| (5) Borrowings from bank to finance acquisition | $670,000 |
| (6) Liability due to Y Co, payable end of 3 years | $1,000,000 |
| Payment due date | 31 Dec 20x6 |
| Interest factor | 5% p.a. |

The financial statements of Topaz Co for the current year ended 31 December 20x6 are shown below:

**Income Statement for year ended 31 December 20x6 and partial Statement of Changes in Equity**

| | |
|---|---|
| Profit before tax | $350,000 |
| Tax | (70,000) |
| Profit after tax | $280,000 |
| Dividends declared | (50,000) |
| Profit retained | $230,000 |
| Retained earnings, 1 Jan 20x6 | 120,000 |
| Retained earnings, 31 Dec 20x6 | $350,000 |

**Statement of Financial Position as at 31 Dec 20x6**

| | |
|---|---|
| Fixed assets, net book value | $590,000 |
| Inventory | 560,000 |
| Accounts receivable | 320,000 |
| Cash | 60,000 |
| | $1,530,000 |

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| | |
|---|---|
| Amount due to P Co | $120,000 |
| Other liabilities | 510,000 |
| Share capital | 450,000 |
| Retained earnings | 350,000 |
| Other comprehensive income (OCI) | 100,000 |
| | $1,530,000 |

***Additional information:***

(a) Shareholders' equity of Topaz at date of acquisition was as follows:

| | |
|---|---|
| Share capital | $450,000 |
| Retained earnings | 100,000 |
| OCI | 6,000 |
| | $556,000 |

(b) The book value of Topaz' identifiable net assets was close to its fair value with the exception of the inventory:

| | |
|---|---|
| Fair value | $160,000 |
| Book value | $120,000 |

The inventory was sold in full in 20x4.

(c) The fair value of non-controlling interests of Topaz as at acquisition date was $188,000.

(d) The balance of other comprehensive income as at 1 January 20x6 was $67,000.

(e) Assume a tax rate of 20%.

***Required:***

1. Prepare the journal entries to record the investment in Topaz, transaction costs, financing transactions and other entries in P Co's books on 2 January 20x4. Recognize unamortized discount separately for future settlements.
2. Record the following transactions in P Co's books for the year ended 31 December 20x6:
   i. Interest expense on long-term loan payable in 20x6;
   ii. Settlement of loan payable on 31 December 20x6.
3. Explain briefly the reason for not recognizing the possible licensing right as an intangible asset as at 2 January 20x4.
4. Prepare consolidation entries for the year ended 31 December 20x6, with narratives (brief headers) and workings in accordance with IFRS 3 and IFRS 10.
5. Perform an analytical check on the balance of non-controlling interests as at 31 December 20x6, showing the workings clearly.

### P4.15 Goodwill impairment

P Co purchased an 80% interest in S Co, which has two divisions: Paints and Chemicals. Each of these two divisions has cash flows that are independent of each other. Hence, they are deemed as "cash-generating units" in accordance with IAS 36 *Impairment of Assets.* The goodwill paid by P Co for S Co was allocated to the two divisions as shown below. The following table shows the balances as at 31 December 20x5.



| | Paints division | Chemicals division |
|---|---|---|
| Goodwill (P's share) | $ 6,000,000 | $ 7,000,000 |
| Goodwill (non-controlling interests' share) | 1,500,000 | 1,750,000 |
| Book value of identifiable net assets | 30,000,000 | 43,000,000 |
| Unamortized balance of fair value adjustments | 3,000,000 | 5,000,000 |
| Fair value of division | 33,000,000 | 56,000,000 |
| Value in use | 37,000,000 | 50,000,000 |

***Required:***

1. What is the impairment loss on goodwill, if any, that should be recognized by P Co in its consolidated financial statements for the year ended 31 December 20x5?
2. The allocation of goodwill to different cash-generating units may be an arbitrary process. Suggest how the allocation may be carried out meaningfully by P Co.

### P4.16 Impairment of goodwill

P Co obtained control of S Co on 1 July 20x6, which has two divisions: Trading and Manufacturing. Each is a "cash-generating unit" as defined by *IAS 36 Impairment of Assets*. P Co chose to measure non-controlling interests at fair value on acquisition date. Total goodwill recognized was $8,000,000. On acquisition, P Co also recognized internally-generated intangible assets at fair value. P Co has to assess the impairment of goodwill as at 31 December 20x8. The following information was available as at 31 December 20x8.

| | Trading division | Manufacturing division |
|---|---|---|
| Goodwill attributable | $ 3,500,000 | $ 4,500,000 |
| Book value of identifiable net assets | 12,000,000 | 20,000,000 |
| Unamortized balance of intangible assets, after tax | 6,000,000 | 4,500,000 |
| Fair value of division | 15,000,000 | 28,000,000 |
| Value in use | 16,000,000 | 27,000,000 |

***Required:***

1. Determine the impairment loss, if any, for goodwill of the trading division and the manufacturing division as at 31 December 20x8.
2. Determine the impairment loss, if any, for identifiable net assets of the trading division and the manufacturing division as at 31 December 20x8.

### P4.17 Impairment of goodwill

P Co obtained control of S Co on 1 July 20x6, which has two divisions: Trading and Manufacturing. Each is a "cash-generating unit" as defined by *IAS 36 Impairment of Assets*. P Co chose to measure non-controlling interests as a proportion of identifiable net assets on acquisition date. Goodwill attributable to P Co that was recognized was $7,000,000. On acquisition, P Co also recognized internally generated intangible assets at fair value. P Co has to assess the impairment of goodwill as at 31 December 20x8. The following information was available as at 31 December 20x8.



| | Trading division | Manufacturing division |
|---|---|---|
| Goodwill attributable | $ 3,000,000 | $ 4,000,000 |
| Book value of identifiable net assets | 12,000,000 | 20,000,000 |
| Unamortized balance of intangible assets, after tax | 6,000,000 | 4,500,000 |
| Fair value of division | 15,000,000 | 27,000,000 |
| Value in use | 16,000,000 | 28,000,000 |

***Required:***

1. Determine the impairment loss, if any, for goodwill of the trading division and the manufacturing division as at 31 December 20x8.
2. Determine the impairment loss, if any, for identifiable net assets of the trading division and the manufacturing division as at 31 December 20x8.

### P4.18 Accounting for other components of non-controlling interests

On 1 January 20x5, Company A acquired all the 800,000 ordinary shares in Company B for $900,000 in cash. The carrying amount or book value of net assets of Company B as at the date of acquisition amounted to $650,000. The fair value of the net identifiable assets as at 1 January 20x5 is $710,000.

During the year, Company B has also issued share options to its employees. As at the date of acquisition, only one tranche of share options has been issued and the carrying value in the share option reserve in equity amounted

to $70,000. The options have vested in the employees and Company A does not replace them on acquisition. There are also no changes to the terms to the share options with the acquisition by Company A.

Additionally, Company B has 1,000 preference shares outstanding as at 1 January 20x5. These shares, which were issued at par of $1 per share, are classified as equity in accordance with IAS 32 *Financial Instruments: Presentation*. The preference shares do not carry any voting rights and will provide their holders a right to a preferred dividend over the payment of any dividend to the ordinary shareholders. Upon liquidation, the preference shareholders are not entitled to a proportionate share of the net assets. The fair value of these preference shares at the date of acquisition is $5,000.

***Required:***

Calculate the goodwill on acquisition and prepare the accounting entries for the transaction. Ignore tax effects for this transaction.



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[1] In practice, the elimination of investment entry is often described as a "permanent" entry because it recurs in each consolidation process in subsequent periods.

[2] The assumption throughout the text is that dividends are tax-exempt.

[3] Basis of Conclusions, IFRS 10, paragraph BCZ165.

[4] Overvaluation generally indicates impairment loss that should be recognized by the subsidiary under IAS 36 *Impairment of Assets*. However, for illustrative purposes, we assume that the impairment loss is not recognized by the subsidiary.

[5] If these employees, share options are vested and the acquirer does not exchange the awards held by the employees for its share-based payment, they are part of the non-controlling interest in the acquiree and are measured at their market-based measure. If the options are not vested at the date of acquisition, they are measured at their market-based measure as if the acquisition date were the grant date. The amount allocated to the non-controlling interest is based on the ratio of the portion of the vesting period completed to the greater of the total vesting period and the original vesting period of the share-based payment transaction. The balance is allocated to post-combination service, that is, part of the consideration transferred is deemed to be payment for this post-combination service. (IFRS 3 paragraphs B62A and B62B)